

University of Chester

MACHINE LEARNING BASED CUSTOMER CHURN PREDICTION IN TELECOM SECTOR

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Abstract

In an increasingly competitive industry, predicting customer churn has become a key priority for businesses striving to retain their clientele. This research focuses on predicting customer churn for a telecom company in California by applying two machine learning algorithms: Logistic Regression and Random Forest. To identify the most influential factors contributing to churn, the study utilized correlation analysis and Random Forest's feature importance tool. Key features, such as contract type, monthly charges, and tenure, were selected for analysis. Logistic Regression emerged as the more accurate model, achieving an accuracy of 83.23%, while Random Forest, though generally more robust, reached a lower accuracy of 74.31%. Additionally, customer segmentation was performed to group clients based on shared characteristics, enabling the business to better understand different customer profiles. This research highlights the importance of careful feature selection and model choice to enhance the accuracy of churn prediction and improve retention strategies.

Contents

Abstract 2
Acknowledgements 4
1. Introduction 5
2. Literature Review 8
2.1 Customer Retention and Churn Dynamics in the Telecom Sector 8
2.1.1 Importance of Customer Retention and Experience in Business Profitability 8
2.1.2 Understanding Customer Churn 9
2.1.3 Factors Affecting Customer Churn in Telecom Sector 10
2.2 Predictive Analytics and Machine Learning in Customer Churn 11
2.2.1 Traditional Approaches to predict Churn 11
2.2.2 Advanced Machine Learning Algorithms in Churn Prediction 19
2.2.3 Comparative Analysis of Churn Prediction Studies in Telecom Industry 24
3. Methodology 28
3.1 Dataset Description 29
3.2 Data Pre-Processing 34
3.2.1 Importing Libraries 34
3.2.2 Loading the Dataset 35
3.2.3 Handling Missing Values 35
3.2.4 Standardizing Target Variable 39
3.2.5 Encoding Categorical Variables 40
3.3 Exploratory Data Analysis 41
3.3.1 Descriptive Statistics and Visualizations 42
3.3.2 Analysing Correlations between Variables 47
3.4 Feature Selection 51
3.4.1 Feature Selection with Correlation Analysis 52
3.4.2 Feature Selection with Random Forest 54
3.4.3 Utilizing Feature Selection for Customer Segmentation Analysis 56
3.5 Model Implementation 58
3.5.1 Logistic Regression 58
3.5.2 Random Forest 62
3.5.3 Comparative Analysis of Model Performances 64
4. Conclusion 66
References 67

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1. Introduction

Over the last two decades, the telecommunication industry has emerged as one of the major industries in the developed countries. This growth can be attributed to rapid technological advancements and a sudden increase in the number of network operators, resulting in intense competition. As digitalization continues to redefine our daily lives, telecom providers have expanded their role beyond mere connectivity; they have become essential facilitators of everyday communication (Sullivan, 2021).

To thrive in this increasingly competitive marketplace, many telecom companies have focused their efforts on developing effective strategies to address one of the major challenges, churn prediction and management. In telecom sector, customer churn, the process by which customers discontinue their service, poses a significant threat to profitability and growth. In mobile telecommunications this concept is closely related to subscriber churn which refers to the migration of subscribers from one network provide to another. Customers often switch to the other network providers based on several factors such as services, rates, or other attractive incentives offered by the new provider (Wei & Chiu, 2002).

Statistics highlight the critical nature of churn management in telecom business. According to (Liibert, 2021) the average churn rate in the telecom industry is 22% which is significantly higher as compared to 19% in IT and 16% in professional services. This indicates that telecom business loses one fifth of customers each year, making it essential for them to invest in effective churn prediction and management strategies

Furthermore, specific company data provides further insight into churn trends. By the end of the first quarter of Vodafone's financial year 2024/25, the contract churn rate for Vodafone customers in the United Kingdom stood at 13.4% (*Vodafone Contract Churn UK 2014-2021*, 2024). This represents an increase from the previous quarter's rate of 12.9%, but a 5.63% decrease compared to the same quarter in the previous year, which was 14.2%. Overall, the contract churn rate at Vodafone UK has been decreasing steadily since 2014. These fluctuations in churn rates at a leading telecom company like Vodafone highlight the dynamic nature of customer retention challenges and the importance of continual improvement in churn prediction techniques. (*Vodafone Contract Churn UK 2014-2021*, 2024)

Kotler (2020) stated that American Communication Service Providers (CSPs) pay a substantial amount of $315 to acquire a new subscriber, making the telecom industry one of the most expensive industries when it comes to customer acquisition cost (CAC). CSPs incur high customer acquisition costs due to offering attractive service bundles, competitive pricing, and substantial marketing efforts to entice subscribers from competitors. Additionally, investments in infrastructure and advanced technology, like mobile-Wi-Fi integration, further drive up these costs. This high cost underscores the financial burden of acquiring new customers as compared to retaining the existing ones. Acquiring new subscribers often involves enticing them away from other service providers, which requires additional incentives to persuade them to switch. Based on this evidence, it is more financially sensible for organizations to focus on retaining their existing customers rather than continually seeking new ones.

The challenge of churn prediction lies in the complexity of customer behaviors and the dynamic nature of the market. Traditional methodologies such as basic statistical analysis and heuristic approaches are no longer used by big businesses to predict customer churn due to their inability to capture the multifaceted factors influencing customer decisions. According to (Sonune et al., 2024) predictive analysis and machine learning algorithms have proven to be robust tools, offering the telecom industry new opportunities to enhance churn prediction accuracy. These powerful tools can help implement specific customer retention strategies by analyzing historical customer data, thereby gaining valuable insights into customer behavior, preferences, and more.

The primary objectives of this dissertation are to explore the efficacy of various machine learning models in predicting customer churn within the telecom industry and to identify the most effective approaches for improving churn management strategies. The research aims to contribute to the existing body of knowledge by providing insights into the practical applications of machine learning in the telecom sector and offering recommendations for future research. This study's significance lies in its potential to help telecom companies enhance their customer retention strategies, thereby reducing the high costs associated with customer acquisition. By leveraging large datasets and sophisticated algorithms, telecom companies can tailor their retention efforts to address the specific needs and behaviors of their customers, ultimately leading to improved profitability and growth.

The scope of this research includes the exploration and application of various machine learning models to predict customer churn within the telecom industry. This research will utilize customer churn data for a fictional telecom company in California sourced through Maven Analytics. The research will compare the effectiveness of different machine learning techniques such as logistic regression, decision trees, random forests, support vector machines, and neural networks in predicting churn. However, the research is limited by the availability and quality of data from the telecom company. Additionally, the study focuses primarily on predictive accuracy and does not delve deeply into the implementation of retention strategies based on the predictions. The generalizability of the findings may also be limited to the specific telecom context and geographic region from which the data is sourced.

The dissertation is structured as follows:

- The Literature review chapter takes a deep dive into the current research on churn prediction and machine learning techniques, giving an overview of the foundation upon which the study builds.
- The Methodology chapter shifts the focus to describing the dataset, the steps taken for data preprocessing, exploratory data analysis, and the analytical methods used. Additionally, the chapter will present the outcomes of the models applied in the study.
- Finally, the conclusion is the final chapter that wraps up by summarizing the key findings, exploring the broader implications of the results, and suggesting directions for future research.

By examining the pressing issue of customer churn in the telecom industry through the application of machine learning, this dissertation aims to provide meaningful insights for both academic research and practical industry solutions.

2. Literature Review

2.1 Customer Retention and Churn Dynamics in the Telecom Sector

2.1.1 Importance of Customer Retention and Experience in Business Profitability

Customers are the prime source of revenue and a crucial asset for any industry. Given this fact, companies have been investing significant time and efforts into providing positive customer experiences. A survey conducted by Stattin (2019), which included responses from 1,920 business professionals, revealed that customer experience is the top priority for businesses.

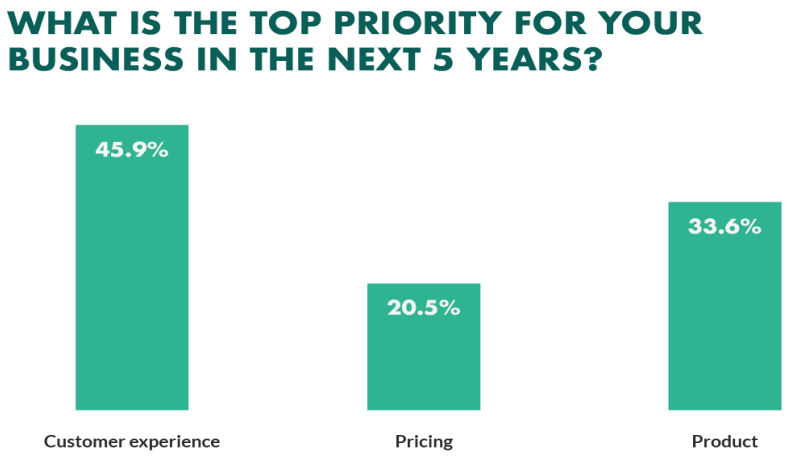


Figure1 showing Top Priority for Business by Stattin (2019)

This focus on customer experience is not without reason. According to Fanderl (2023) the likelihood of retaining satisfied customers is 20-30% higher than retaining those who are not satisfied. While many companies focus on acquiring new customers, it has been evident that retaining existing customers leads to better financial results and lays the foundation for future growth. McKinsey & Company states:

"To make up for the loss of one existing customer, companies have to acquire three new customers."

This emphasizes the importance of customer retention as a critical business strategy. An article published by Forbes states that a mere 5% increase in customer retention

can result in a substantial profit growth ranging from 25% to 95% over time (Kumar, 2024). Therefore, by prioritizing customer satisfaction and experience, businesses not only improve retention rates but also enhance overall profitability and sustainable growth. Additionally, understanding and predicting customer churn has become quite essential for companies to effectively manage and improve their customer retention strategies.

2.1.2 Understanding Customer Churn

2.1.2.1 Customer Churn in Telecom Sector

Customer churn has emerged as a significant concern across various industries, particularly in the telecommunications sector which is grappling with this daunting challenge (Jain et al., 2020).

(Wagh et al., 2024) define customer churn as a phenomenon where existing customers discontinue their service with a provider over a given period, with these customers being referred to as 'churners'. The primary goal of churn analysis is to predict these churners early and understand their reason for leaving. Early identification allows companies to focus on two strategies to reduce churn:

A. Proactively address customer issues to increase satisfaction and retention.
B. Identify the characteristics of groups of customers most likely to churn and target them with specific retention strategies.

An article published by Oliver Wyman (Amiot et al., n.d.), reveals that in the European mobile service market, a significant proportion of customers are inclined to switch providers.

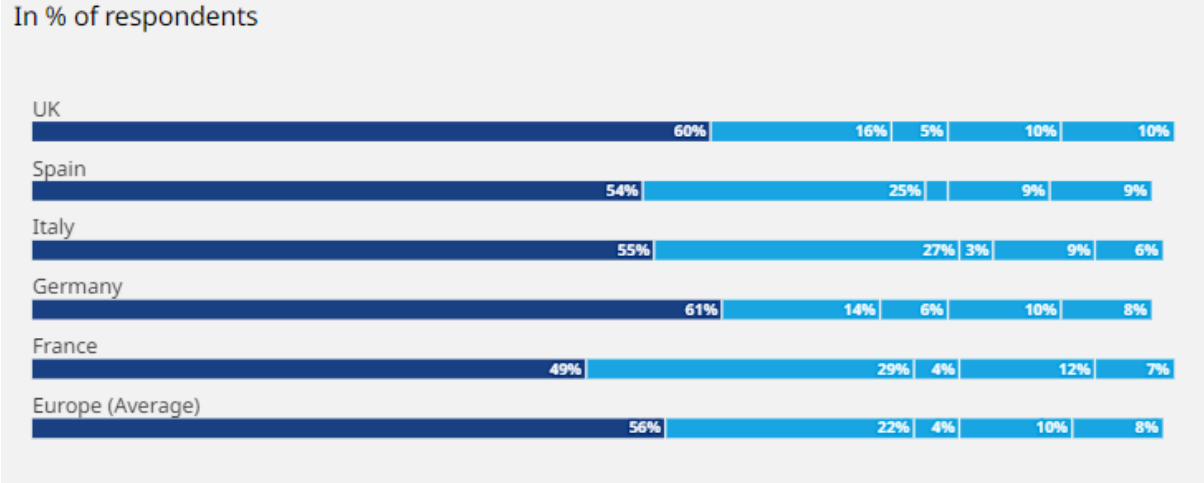


Figure 2 representing Intention to switch mobile phone service provider, by country (Amiot et al., n.d.)

For instance, 60% of UK respondents, 55% of Italians, and 54% of Spanish consumers have expressed their intention to switch mobile service providers, highlighting the competitive nature of the telecom sector in Europe. This substantial willingness to switch among European consumers underscores the importance of accurate churn prediction in retaining customers.

To address this issue, telecom companies are investing in advanced machine learning predictive models to identify potential churners. This enables them to target these customers with additional benefits, plans and incentives, rather than offering them to regular customers, to retain them. However, this approach can result in a significant loss if the model fails to accurately evaluate churn measures.

Inaccurate identification of churners in predictive model can lead to significant loss for telecom companies. When companies misallocate retention efforts to customers who would not have left, they waste valuable resources and miss the opportunity to focus on those who truly require intervention. This can result in decreased customer satisfaction and missed opportunities for long-term retention. (Wagh et al., 2023).

This underscores the importance of effective churn prediction and management in maintaining profitability and ensuring long-term growth.

2.1.3 Factors Affecting Customer Churn in Telecom Sector

Several factors contribute to customer churn in the telecom sector, each playing a crucial role in influencing a subscriber's decision to switch providers. In their study, (S et al.,

2014) identified multiple key factors contributing to customer churn, focusing on Network Quality, Call Facilities, and Internet Services as the most significant. Using a survey of 750 respondents—categorized as students, professionals, and businesspersons—the study analysed the data through the ID3 Decision Tree algorithm in WEKA to determine the highest impact factors. The results indicated that poor Network Quality was a critical factor across all segments, while students were also heavily influenced by Booster Facilities. In contrast, professionals prioritized Internet Facilities, and businesspersons were significantly affected by both Call Facilities and Booster Services.

These findings underline the necessity for telecom providers to enhance service quality, especially in network coverage and additional services like boosters and internet access, to effectively reduce churn in different customer groups

This literature review aims to explore the evolution of churn prediction methods, with a particular focus on the application of machine learning techniques.

2.2 Predictive Analytics and Machine Learning in Customer Churn

2.2.1 Traditional Approaches to predict Churn

Customer churn prediction has been a critical area of focus for the telecom industry, as it directly impacts profitability and customer retention. Over the years, traditional approaches to churn prediction have laid the groundwork for more advanced techniques, despite certain limitations. These traditional methods include statistical models, rule-based systems, and early data mining techniques.

2.2.1.1 Statistical Methods

Statistical methods have been among the earliest tools used in churn prediction and are valued for their simplicity, interpretability and ease of implementation.

2.2.1.1.1 Logistic Regression

A widely utilized statistical technique, is particularly effective in modelling the probability of a binary outcome. According to IBM (2024) Logistic Regression is often employed for classification and predictive analysis, where the model predicts the probability of a binary event – such as customer churn, based on a given dataset. The application of Logistic Regression in Churn Prediction has been extensively documented in literature. For instance, (Coussement & Van den Poel, 2008)

conducted a study using data from a Belgian newspaper company to predict customer churn. They compared several models, including Logistic Regression, Support Vector Machines (SVMs), and Random Forests. While Logistic Regression was used for its simplicity and interpretability, the study found that SVMs and Random Forests, particularly with optimized parameters, outperformed it in predictive accuracy.

A key takeaway was the importance of feature engineering and selection in training the model. The researchers used Random Forest to rank the most effective churn predictors. Unlike other studies, this research found that the most significant churn predictors were related to subscription behaviour, while monetary features and transaction frequency, often emphasized in other research, were not among the top 10 drivers. Instead, client/company interaction variables played a significant role, and socio-demographics, aside from age, proved less influential in explaining churn.

The research highlighted the effectiveness of these advanced models but also noted limitations, such as the specificity of the dataset and the focus on technical performance over practical application. The study focused on a specific dataset from a Belgian newspaper company, which may limit the generalizability of findings to other industries or regions. Furthermore, although logistic regression was used as a benchmark, the study did not delve deeply into optimizing this model and overlooked opportunities to improve the Logistic Regression model through techniques like regularization or feature engineering. Therefore, the study revealed that while Logistic Regression is useful, more complex models may offer better results in noisy, real-world datasets.

Similarly, (Jain et al., 2020a) study utilized a dataset from an American telecommunication company named Orange containing data for 3,333 customers including 20 attributes, reduced to 11 after data cleaning. The target variable, "churn," indicates whether a customer has churned or not, with 483 instances of churners in the dataset.
The research applied 2 machine learning techniques for churn prediction: Logistic Regression and Logit Boost. Several performance metrics, including Kappa Statistic, Mean Absolute Error, Mean Square Error were used to evaluate the model performance. The research also discussed briefly about data cleaning phase. During

the data preparation stage, certain attributes, such as state and date of birth (DOB), were excluded due to their limited predictive power or excessive null values. Additionally, fields that contributed the same information were removed to improve model efficiency and reduce database size.

In terms of model choice, the study also provided a visual comparison between linear and logistic regression models to highlight the appropriateness of using Logistic Regression for binary classification problems, such as churn prediction.

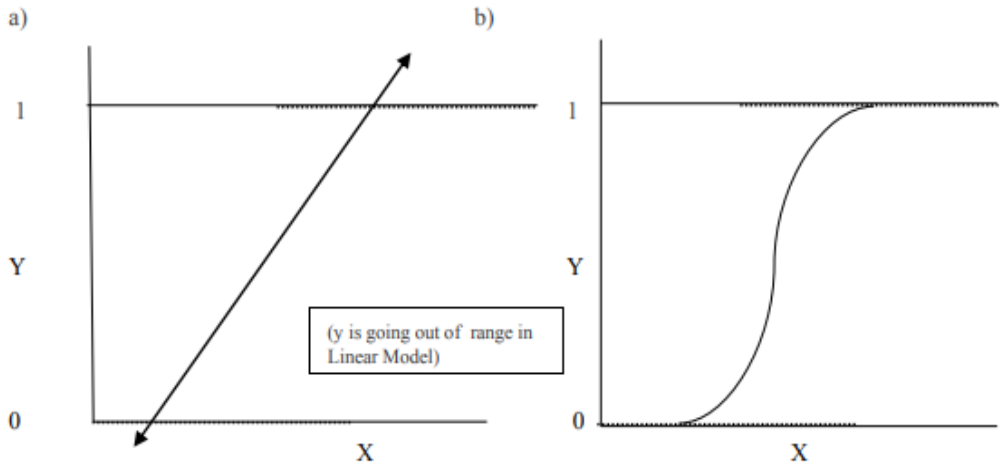


Fig. 1. (a) Linear Model; (b) Logistic Regression Model.

Figure 3 showing Comparison between Linear and Logistic Regression Model by Jain et al. (2020a)

The authors provided a visual comparison Figure 3 showing how a linear model (Fig. 1a) can produce predictions that extend beyond the range of valid probabilities (0 to 1), making it unsuitable for binary outcomes. In contrast, the logistic regression model (Figure 1b) naturally constrains the output within the [0, 1] interval through the S-shaped logistic function, ensuring that predictions are interpretable as probabilities. This comparison underscores the importance of choosing a model that aligns with the nature of the prediction problem, which in this case validates the use of Logistic Regression over a simple linear model for predicting customer churn. The results after dividing the dataset into training and testing sets show that both the techniques performed quite similarly, with Logistic Regression achieving an accuracy of **85.24%**, while Logit Boost had an accuracy of **85.18%**.

The author concluded that both Logistic Regression and Logit Boost performed well in predicting customer churn, with Logit Boost offering a slight edge due to its

iterative improvement of prediction accuracy. Despite the impressive performance of both models, it is noteworthy that the study did not address feature engineering. The absence of discussion around feature engineering suggests a potential area for enhancements, as selecting accurate features to train the model can effectively capture more complex patterns in customer behaviour, leading to better predictive accuracy.

2.2.1.1.2 Survival Analysis

"Survival analysis, which is an important subfield of statistics, provides various mechanisms to handle such censored data problems when they arise in modeling complex data. This is also referred to as time-to-event data when modeling a particular event of interest is the main objective and is very common in real-world application domains." (Wang et al., 2019)

As highlighted by Wang et al., survival analysis is not only valuable in determining whether customers are likely to churn in the telecom industry but also in predicting when they are likely to churn. This predictive approach is crucial for developing targeted retention strategies that can be implemented just before the risk of churn increases, thereby enhancing the effectiveness of retention efforts.

The research by (Havrylovych & Nataliia Kuznietsova, n.d.) illustrates the application of survival analysis in this domain by developing algorithms based on survival functions to forecast churn. They employed both parametric and semi-parametric survival models, including the Cox Proportional Hazards Model, Weibull, and Log-normal models, to predict churn risk and timing.

Covariate name	Weibull		LogNormal		Cox Model	
	Exp(coef)	p-value	Exp(coef)	p-value	Exp(coef)	p-value
Partner	1.67		1.75		0.59	
Dependents	0.91	0.19	0.92	0.23	1.05	0.49
MultipleLines	1.74		1.87		0.59	
InternetService	0.39		0.41		2.38	
OnlineSecurity	2.14		2.17		0.49	
OnlineBackup	2.01		2.08		0.49	
DeviceProtection	1.47		1.53		0.68	
TechSupport	1.72		1.65		0.62	
StreamingTV	1.18	0.01	1.26		0.86	0.01
StreamingMovies	1.29		1.28		0.78	
Contract	1.14		1.13		0.87	
PaperlessBilling	0.81		0.85	0.01	1.21	
MonthlyCharges	0.69		0.70		1.43	
automatic_payment	1.95		2.01		0.54	
Concordance	0.87		0.87		0.87	
Log-likelihood	-8921.22		-8845.61		-13889.35	

Figure 4 showing results of Weibull, Lognormal and Cox Models by (Havrylovych & Nataliia Kuznietsova, n.d.)

As seen in Figure 4, the Log-normal model provided the best performance with a higher log-likelihood value (-8845.61), indicating a better fit compared to the Cox model (-13889.35) and the Weibull model (-8921.22). The Log-normal model also showed strong statistical significance across several covariates, suggesting its robustness in predicting churn. However, while the Log-normal model outperformed others in statistical criteria, the Cox model remains highly valued for its ability to handle multiple covariates and censored data, as well as its interpretability.

Like other machine learning models, the success of these survival modes is also heavily dependent on effective feature selection. In their study, the authors identified critical features that significantly impact churn prediction. Key features included customer account information (e.g., tenure, contract type, payment method), service utilization (e.g., online security, tech support), and demographic variables (e.g., gender, age, partner status). Notably, variables such as Online Security, Partner, and Internet Service were pivotal in predicting customer loyalty or churn. For instance, customers without online security exhibited a higher propensity to churn, highlighting how data safety significantly influences customer retention.

The Cox Proportional Hazards Model was applied to predict the timing of customer churn. The research also explored other survival models, such as the Weibull and

Log-normal models, and concluded that while the Cox model was effective, the Log-normal model offered better performance based on statistical criteria like log-likelihood value. However, the Cox model's ability to incorporate multiple covariates and handle censored data still makes it a valuable tool in the l of churn prediction techniques.

The Cox Proportional Hazards Model remains a robust method for survival analysis in churn prediction, offering telecom companies the ability to estimate not just who is likely to churn, but when. This temporal dimension is crucial for implementing timely and cost-effective interventions. Despite its limitations, the Cox model's flexibility and applicability in handling censored data and incorporating various customer characteristics make it an essential component of churn prediction strategies.

2.2.1.2 Early Data Mining Techniques

Data mining techniques emerged as a crucial development in the field of customer churn prediction, bridging the gap between simple statistical methods and more complex machine learning models.

(Thearling & Com, n.d.) defined Data Mining as ‘The automated extraction of predictive information from large databases’, a crucial tool for revealing hidden patterns and relationships within vast datasets, enabling business to make informed decisions and predict future trends with greater accuracy.

Lejeune (2001) emphasized that, as data continues to grow in volume and complexity, there has been an increasing reliance on data mining methods. Lejeune pointed out that data mining not only enhances the understanding of existing data but also provides predictive insights that are essential for timely decision-making in competitive markets. This is particularly important as businesses face the challenge of reducing the time between the emergence of trends and their detection, ensuring they can respond swiftly to changes in the market.

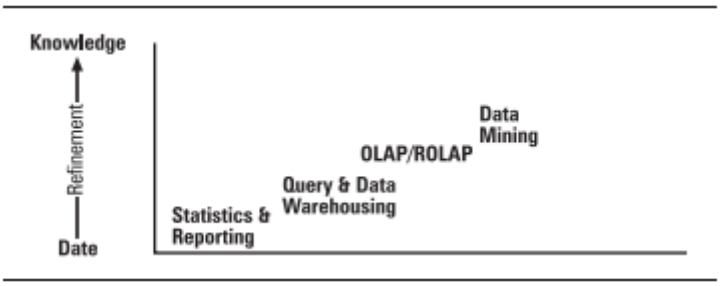


Figure 5 demonstrating evolution of techniques & quest for information by Lejeune (2001)

Figure 5 illustrates the evolution in the quest for information, highlighting the progression from basic statistics and reporting to more advanced techniques such as data mining.

In the telecom industry, early data mining approaches played a significant role in identifying at-risk customers and informing retention strategies.

2.2.1.2.1 Decision Trees

Decision trees were among the earliest and most popular data mining techniques used for churn prediction due to their simplicity and interpretability.

The research by (Bin et al., 2007) provides valuable insights into the utility of decision trees as an effective tool in data mining, particularly for classification and prediction tasks. Their study highlights the nonparametric nature of decision trees, which allows them to function without requiring assumptions about data distributions. They are constructed in a tree structure, with branches extending to terminal nodes that represent classification outcomes. Decision trees are robust against noise and relatively easy to interpret, especially when overfitting prevention techniques are applied. These qualities make decision trees one of the most used methods in data mining, as they effectively uncover important patterns and relationships within complex datasets.

In the context of churn prediction in the telecom sector, decision trees have been applied with notable success. A prime example is the study by Ahmad et al. (2019), which explores the development of a churn prediction model using various machine learning techniques, including decision trees, to analyse customer behaviour and predict the likelihood of churn. The study utilized a nine-month dataset from SyriaTel, encompassing customer demographics, usage patterns, and social network analysis (SNA) features. Several algorithms, such as Decision Trees, Random Forest, GBM,

and XGBoost, were tested, with XGBoost delivering the highest accuracy.

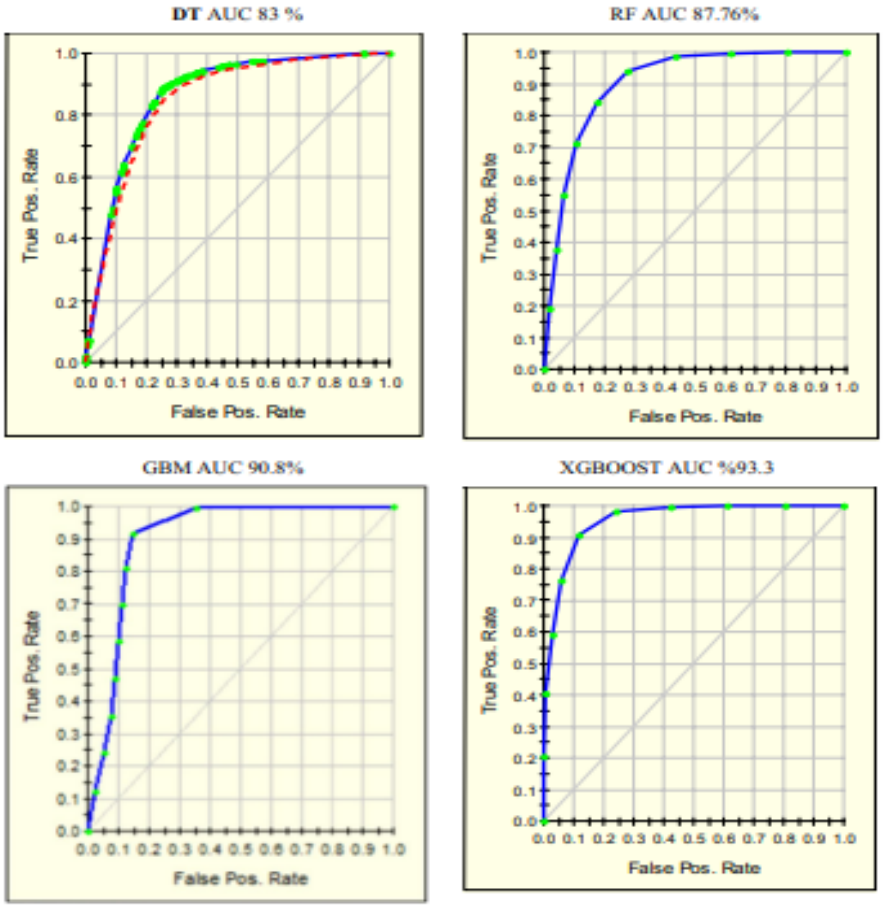


Figure 6 representing characteristic curve for each classification algorithm by (Ahmad et al., 2019)

This performance is reflected in the ROC curves, where XGBoost achieved the highest AUC score of 93.3%.

A vital aspect of this research was featuring selection and transformation. Given the complexity of customer behaviour, a comprehensive dataset was analysed, totalling over 70 terabytes of data collected over nine months. The process involved aggregating numerical data (e.g average, counts) and counting distinct categorical values. Graph Frame Library and Spark engine were mainly utilized to compute the features reflecting social activities through SMS and calls. Statistical features from call detail records (CDRs) included metrics like average monthly calls, internet usage, and subscription counts. Daily, weekly, and monthly data aggregation significantly expanded the feature set, which also incorporated customer complaints metrics.

Additionally, device-related features and customer interactions with competitors were extracted

Furthermore, in the feature selection phase unique identifiers and features with

excessive values were removed, resulting in a refined dataset of 2000 features. The key features identified as a part of feature selection process included average monthly calls, internet usage statistics, customer complaint metrics, and social network analysis.

The study concluded that these models, especially when incorporating decision trees and SNA, are effective in predicting customer churn, ultimately helping the telecom company reduce churn rates and increase revenue.

Although, the study focused on accuracy as the primary metric, but it could have also considered other performance metrics like precision, recall, and the impact of churn prediction on customer retention strategies to provide a more holistic evaluation of the models.

2.2.2 Advanced Machine Learning Algorithms in Churn Prediction

An effective approach to churn modeling involves two main objectives: first, forecasting whether a particular customer is at risk of leaving, and second, understanding the underlying reasons for their potential departure. While most research has traditionally focused on the former, this approach alone is insufficient for fully understanding customer behavior. To effectively reduce churn, it is essential for telecom providers to not only identify which customers might leave but also to comprehend the factors driving this behavior, enabling them to target at-risk customers with tailored retention campaigns (Sumana Sharma Poudel et al., 2024).

In recent years, predicting and preventing customer churn has become a critical focus for companies, driven by the vast amounts of business data accumulated through automated operations. This data provides a solid foundation for the application of advanced machine learning algorithms, which have significantly transformed the approach to customer churn prediction (Zhu et al., 2009). While numerous models have been explored in this field, including Decision Trees, Artificial Neural Networks, and Logistic Regression, no single method consistently outperforms others across all metrics. Nevertheless, models like Random Forests, Support Vector Machines (SVMs), and Logistic Regression, as employed in this study, offer a balance of accuracy and flexibility, enabling businesses to develop more effective churn management strategies.

2.2.2.1 Random Forest

Random Forest has become a powerful tool in customer churn prediction, particularly valued for its effectiveness in handling complex, application-specific problems within customer relationship management. This machine learning technique excels at identifying and ranking key churn indicators, which are crucial for accurately predicting and preventing customer attrition (Ullah et al., 2019). In the context of the telecommunications industry, Random Forest is especially advantageous due to its ability to manage large datasets, address non-linear relationships, and handle correlated variables commonly found in customer behaviour data.

One notable study by (Sunil, Anjali A et al., 2023) applied the Random Forest classifier to predict customer churn using a dataset from an Iranian telecom company, comprising 3,150 customer records with 13 attributes. After data preprocessing, the Random Forest model achieved a high accuracy of 98.31%, outperforming other machine learning models. The study identified key churn predictors like customer value and complaint frequency, offering actionable insights for telecom companies to enhance retention strategies.

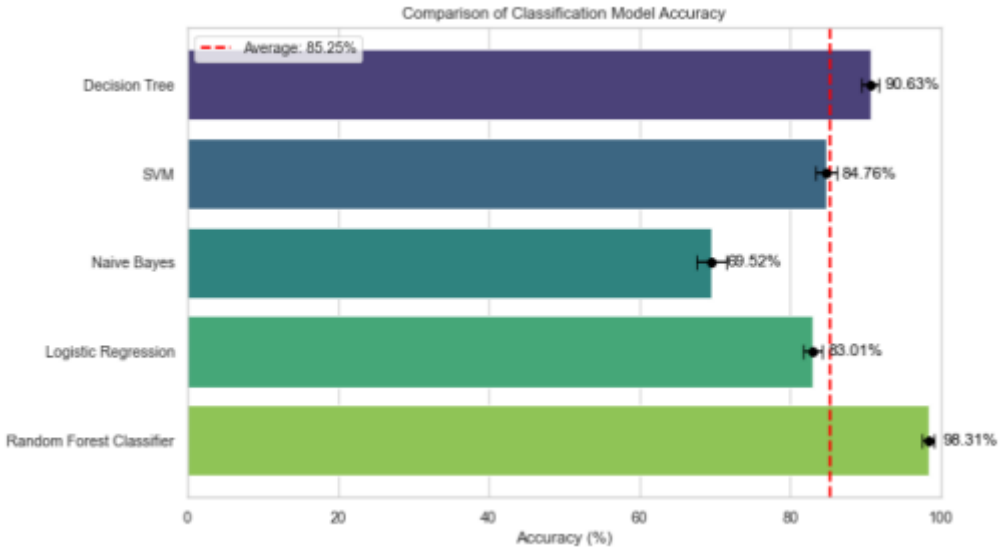


Figure 7 representing Comparison of Classification Models Accuracy by (Sunil, Anjali A et al., 2023)

The finding of this study is visually summarized in a comparison chart, which illustrates how the Random Forest classifier outperformed other models like Decision Trees, SVM, and Logistic Regression, significantly exceeding the average model accuracy of 85.25%. The research also showed that while deep learning models perform well, Random Forest remains highly competitive, particularly when used in hybrid approaches, making it a valuable tool in churn prediction.

In contrast, (Natarajan et al., 2024) proposed an advanced version of the Random Forest algorithm known as Modified Random Forest (MRF) to improve the prediction of customer churn. Their study focused on addressing issues like overfitting and regression errors that are common in churn and upsell forecasting. Using a dataset from the Orange Telecom (2009 KDD Cup), they demonstrated that the MRF model outperformed the standard Random Forest in terms of accuracy and AUC (Area Under Curve), achieving an impressive accuracy of 99.53% compared to the 98.58% of the conventional Random Forest.

These studies collectively underscore the versatility and effectiveness of Random Forest algorithms in customer churn prediction. Whether using the standard approach or an enhanced version like MRF, Random Forest continues to demonstrate superior accuracy and reliability, making it a critical tool for improving customer retention strategies in the telecommunications industry.

2.2.2.2 Support Vector Machine

Support Vector Machines (SVMs) have emerged as a powerful and versatile tool in the realm of advanced machine learning algorithms, particularly in the context of customer churn prediction.

(Brandusoiu & Toderean, 2013) present a comprehensive study on using SVMs to predict customer churn in the telecommunications sector. The study employed a dataset containing 3,333 customer records with 19 attributes. The exploratory data analysis highlighted that the dataset is imbalanced, as only 14% of customers are churners, leading to bias towards the non-churn group. To address these various techniques such as SMOTE and Tomek Links were applied in different stages of the feature selection to balance the dataset.

Prior to feature selection, data cleaning excluded irrelevant variables like "State" and "Area Code" due to inconsistencies, as the area codes were limited to California, reducing noise in the model. For feature selection, two effective techniques—Filter method and Sequential Feature Selection (SFS)—were employed to identify the most relevant features by removing irrelevant and redundant ones. The table in **Figure 8** shows that features such as “Account length,” “International plan,” “Voice mail plan,” and “Total day charge” were consistently selected by both methods. These features offer direct insights into the customer’s relationship with the telecom provider. The features were further refined, and in some cases, variables like “Total intl charges” and “Total intl minutes”

were also removed. Depending on the selection and balancing techniques used, the final number of selected features ranged between 9 and 14.

These selected features offer quick insight into which customers are more likely to churn. For instance, those with international plans or who heavily use day and night minutes were consistently flagged as higher churn risks. Frequent customer service calls also stood out as a sign of possible dissatisfaction, often leading to churn. In contrast, features like voicemail messages or night calls were less relevant, as they weren't consistently chosen in the selection process.

No.	Variables	Variable description	Variable types
1	State	51 states of the US	Categorical
2	Account length	Active duration of accounts	Integer
3	Area code	Code of areas	Categorical
4	International plan	1 = use service, 2 = not use service	Boolean
5	Voice mail plan	1 = use service, 2 = not use service	Boolean
6	Number vmail messages	Amount of voice mail messages	Integer
7	Total day minutes	Total day call minutes clients have used	Continuous
8	Total day calls	Total amount of day calls	Integer
9	Total day charge	Total day fee	Continuous
10	Total eve minutes	Total call minutes clients have used in the evening	Continuous
11	Total eve calls	Total amount of evening calls	Integer
12	Total eve charge	Total evening fee	Continuous
13	Total night minutes	Total night call minutes clients have used	Continuous
14	Total night calls	Total amount of night calls	Integer
15	Total night charge	Total night fee	Continuous
16	Total intl minutes	Total call minutes clients have used for international calls	Continuous
17	Total intl calls	Total amount of international calls	Integer
18	Total intl charge	Total International fee	Continuous
19	Customer service calls	Amount of customer service calls made	Integer
20	Churn	1 = churner, 0 = non-churner	Boolean

https://doi.org/10.1371/journal.pone.0267935.t002

Figure 8: Summary of Feature Selection Process (Brandusoiu & Toderean, 2013)

The SVM algorithm was then implemented using four different kernel functions: Radial Basis Function (RBF), Linear, Polynomial, and Sigmoid. These kernels transform the input data into high-dimensional feature spaces, enabling the separation of data points even when they are not linearly separable in the original space.

CONFUSION MATRIX FOR SVM

Kernel Type	*Observed (Churn?)*	*Predicted*		
		No	Yes	% correct
RBF	No	498	79	86.31%
	Yes	19	86	*81.90%*
	Overall %	75.8%	24.2%	85.63%
LIN	No	448	129	77.64%
	Yes	22	83	*79.05%*
	Overall %	68.9%	31.1%	77.86%
POL	No	519	58	89.95%
	Yes	20	85	*80.95%*
	Overall %	79%	21%	88.56%
SIG	No	406	171	70.36%
	Yes	30	75	*71.43%*
	Overall %	63.9%	36.1%	70.53%

Figure 9 showing Confusion Matrix for SVM by (Brandusoiu & Toderean, 2013)

As illustrated in the confusion matrix, the SVM model with the Polynomial kernel achieved the highest accuracy at 88.56%, correctly predicting 89.95% of non-churners and 80.95% of churners. The RBF kernel also performed well, with an overall accuracy of 85.63%, particularly excelling in identifying churners with an accuracy of 81.90%. On the other hand, the Sigmoid kernel showed the lowest performance, achieving only 70.53% accuracy. Gain measures further validated that Polynomial and RBF kernels significantly improved prediction accuracy, especially for the top 20% of subscribers at risk of churning. How ever, this study mainly shed light on comparing different kernels without delving deeply into the optimization of SVM parameters.

In contrast, the study by (Rodan et al., 2014) takes a more focused approach by applying SVM specifically to a local telecommunications company, addressing the class imbalance problem inherent in churn data. This study also showcased the feature engineering process in depth, identifying key attributes influencing customer churn, such as calling minutes and 3G service usage. The data cleaning stage involved thoroughly processing a dataset of 5000 customers and normalizing features like consumption and SMS fees, and reducing bias. After the feature selection process, 11 releavnt features were chosen for further training and testing. Churners were identified by behaviors like reduced core service usage, higher international charges, and erratic service patterns, while loyal customers showed consistent usage, adoption of new services, and higher spending. This process enabled the model to effectively differentiate between churners and loyal customers. The authors implemented a grid search technique to fine-tune key parameters,

such as the regularization constant CCC and the RBF kernel parameter γ\gammaγ, achieving an impressive accuracy of 98.7%. This tuning process was crucial in maximizing the model's predictive power, which surpassed the accuracy reported by (Brandusoiu & Toderean, 2013). Moreover, (Rodan et al., 2014) emphasized the importance of using a customized evaluation metric to address the class imbalance issue, a factor that significantly impacts the reliability of churn predictions.

2.2.3 Comparative Analysis of Churn Prediction Studies in Telecom Industry

The table 1 below provides an overview of various studies discussed in literature review section on predicting customer churn in the telecommunication sector. It highlights different approaches and methods, such as logistic regression, random forests, and support vector machines, showcasing their effectiveness in terms of predictive accuracy. Each study reveals not only key findings but also common challenges, such as the need for better feature engineering and a wider exploration of performance metrics beyond just accuracy.

Table 1: Comparative Analysis of Churn Prediction Studies in the Telecom Industry

Authors	**Dataset**	**Method/Models**	**Key Findings**	**Limitations**	**Gaps Addressed**
Coussement and Van den Poel (2008)	Belgian Newspaper Company Data	Logistic Regression, SVM, Random Forest	• SVM and Random Forest outperformed Logistic Regression in predictive accuracy. • Highlighted client interaction variables	• Focused on specific dataset and did not optimize Logistic Regression effectively.	Optimized Logistic Regression with feature Engineering and tests on broader datasets to enhance applicability.
(Jain et al., 2020a)	Orange Telecom dataset consisting of 3,333 customers	Logistic Regression, Logit Boost	• Both the models performed similarly (accuracy approximately	• Did not address feature engineering; limited discussion	Incorporates feature engineering and evaluates multiple performance metrics beyond

			equals to 85%). • Logit Boost had a slight edge due its iterative improvement of prediction accuracy.	on model performance metrics.	accuracy.
Havrylovych and Nataliia Kuznietsova (n.d.)	Various Telecom customers	Survival Analysis (Cox Proportional Hazards, Weibull)	• The Log-normal model showed the best performance in churn prediction. • Key predictors for identifying customer churn included service utilization, customer account information and demographic variables.	• Dataset acquisition and details about dataset (no. of customers and features/attributes) are not discussed. • Limited discussion on practical applications; Cox model was not optimized for specific use cases	Applies enhanced survival analysis techniques to identify predictors and their implications for retention strategies.
Ahmad et al. (2019)	SyriaTel dataset, over 70 TB of data	Decision Trees, Random Forest, GBM, XGBoost	• XGBoost achieved highest accuracy (AUC of 93.3%. • The study emphasized in depth on feature selection and transformation processes.	• Focused mainly on accuracy; overlooked other metrics like F1 score. precision and recall.	Evaluates multiple metrics including precision and recall for a more holistic performance assessment of models.

			• Key Features identified to predict churn included: average monthly calls, internet usage statistics, customer complaint metrics, and social network analysis.		
Sunil, Anjali A et al. (2023)	Iranian telecom company dataset consisting of 3,150 records and 13 attributes	Random Forest, Logistic Regression, Naïve Bayes, SVM and Decision Trees	• Random Forest achieved a high accuracy (98.31%). • The key attributes identified are customer value and complaint frequency.	• The model was able to achieve high accuracy as the dataset is very small. • Limited exploration of hybrid approaches . • The study did not discuss in depth about exploratory data analysis and feature selection methods utilized.	Explore effective feature selection techniques to improve predictive power and applicability on larger datasets.
Natarajan et al. (2024)	Orange Telecom dataset (2009 KDD Cup)	Modified Random Forest (MRF)	• MRF outperformed standard Random Forest in accuracy (99.53% vs	• The study did not discuss data loading, pre-processing	In depth research on data pre-processing, feature selection and exploratory

			98.58%); focused on addressing overfitting issues.	and feature selection process in depth.	data analysis techniques.
Brandusoiu and Toderean (2013)	3,333 customer records with 19 attributes	Support Vector Machines (SVM)	• SVM with Polynomial kernel achieved high accuracy (88.56%); highlighted class imbalance and feature selection techniques. • Extensive feature selection techniqes were applied and the key features included: day charge, evening minutes, evening charge, night minutes and night calls.	• Did not delve deeply into SVM parameter optimization and focused mainly on kernel comparison.	Evaluates multiple metrics including precision and recall for a more holistic performance assessment of models.
Rodan et al. (2014)	Local telecommunications company dataset, 5,000 records	Support Vector Machines (SVM)	• Identified key attributes influencing churn. • Achieved 98.7% accuracy through parameter tuning and normalization.	• The study did not discuss in depth about exploratory data analysis and feature selection methods utilized.	• Worked on customer segmentation and customer profiling

3. Methodology

The methodology of this research explores the stages of building a customer churn prediction model using a structured, multi-step approach.

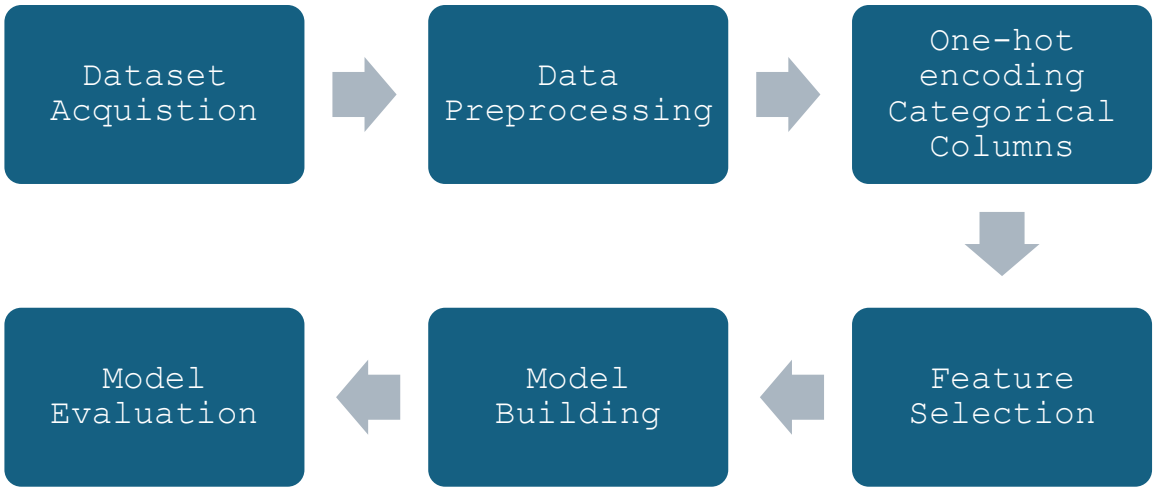


Figure 10: Proposed Model for Customer Churn Prediction

As shown in Figure 10, the first step in building this customer churn prediction model is sourcing the dataset from Maven Analytics website followed by importing necessary libraries and loading the dataset in the working environment. Following this, data preprocessing was performed to ensure the dataset's quality by handling missing values and removing unnecessary columns. Categorical variables were one-hot encoded, while numerical features were standardized to enhance model performance. In addition, correlation analysis and feature importance techniques using a Random Forest classifier were employed to identify significant features for churn prediction. Subsequently, different machine learning algorithms, including Logistic Regression and Random Forest, were applied to classify customers based on their churn status. The models were evaluated using various performance metrics, such as accuracy and classification reports, to assess their effectiveness in predicting customer churn. Finally, visualizations were generated to explore key customer attributes, providing insights into factors affecting churn behaviour.

3.1 Dataset Description

The dataset used for this research is a telecom churn dataset obtained from the Maven Analytics website Free Data Sets and Dataset Samples | Maven Analytics (2024). It includes data for 7,043 customers of a telecommunications company based in California, comprising 38 attributes. Figure 11 below provides a detailed description of each feature and its role in the telecom churn dataset.

Column	Description
CustomerID	It is a unique ID that identifies each customer.
Gender	The customer's gender: Male or Female.
Age	The customer's age, in years, as of the end of fiscal quarter Q2 2022.
Married	The customer's marital status which identifies if a customer is married or not: Yes or No.
Number of Dependents	It specifies the number of dependents living with the customer (dependents could be children, parents, grandparents or others)
City	The city where each customer resides primary in California.
Zip Code	The postal code of the customer's primary residence.
Latitude	The geographic latitude of the customer's primary residence.
Longitude	The geographic longitude of the customer's primary residence.
Number of Referrals	It specifies total number of times the customer has referred a friend or a member of family to the company so far.
Tenure in Months	It specifies the total number of months the customer has been with the company as of the end of the after mentioned quarter.
Offer	It indicates the most recent marketing offer accepted by the customer: None, Offer A, Offer B, Offer C, Offer D, or Offer E.
Phone Service	It specifies whether the customer subscribes to home phone services with the company: Yes or No.
Avg Monthly Long-Distance Charges	It indicates the customer's average long-distance charges, calculated to the end of the quarter specified above (if the customer is not subscribed to home phone service, this will be 0).
Multiple Lines	It specifies whether the customer subscribes to multiple telephone lines with the company: Yes or No (if the customer does not have home phone service, this will be No).
Internet Service	It specifies whether the customer has a subscription to Internet service with the company: Yes or No.
Internet Type	It specifies the type of internet connection the customer uses: DSL, Fiber Optic, or Cable (if the customer is not subscribed to internet service, this will be None).
Avg Monthly GB Download	It specifies the customer's average download volume in gigabytes, calculated up to the end of the aforementioned quarter (if the customer is not subscribed to internet service, this will be 0).
Online Security	It specifies whether the customer subscribes to an additional online security service offered by the company: Yes or No (if the customer does not have internet service, this will be No.
Online Backup	It specifies whether the customer subscribes to an additional online backup service offered by the company: Yes or No (if the customer does not have internet service, this will be No)
Device Protection	It specifies if the customer subscribes to an additional device

Plan	protection plan for their Internet equipment provided by the company: Yes or No (if the customer is not subscribed to internet service, this will be No)
Premium Tech Support	It specifies whether the customer subscribes to an additional technical support plan from the company that offers reduced wait times: Yes or No (if the customer does not have internet service, this will be No)
Streaming TV	It specifies whether the customer uses their Internet service to stream television programming from a third-party provider at no extra charge: Yes or No (if the customer does not have internet service, this will be No).
Streaming Movies	It specifies whether the customer uses their Internet service to stream movies from a third-party provider at no extra cost: Yes or No (if the customer does not have internet service, this will be No).
Streaming Music	It specifies whether the customer uses their Internet service to stream music from a third-party provider at no extra charge: Yes or No (if the customer does not have internet service, this will be No).
Unlimited Data	It specifies whether the customer has paid an extra monthly fee for unlimited data downloads/uploads: Yes or No (if the customer does not have internet service, this will be No)
Contract	It specifies the customer's current contract type which could be Month-to-Month, One Year, or Two Year.
Paperless Billing	It specifies whether the customer has opted for paperless billing: Yes or No.
Payment Method	It specifies the method by which the customer pays their bill: Bank Withdrawal, Credit Card, or Mailed Check.
Monthly Charge	It specifies the customer's total monthly charge for all services provided by the company.
Total Charges	It specifies the customer's total charges calculated up to the end of the aforementioned quarter.
Total Refunds	It specifies the customer's total refunds, calculated to the end of the quarter specified above
Total Extra Data Charges	It specifies the customer's total charges for additional data downloads exceeding those included in their plan, as of the end of the aforementioned quarter.
Total Long-Distance Charges	It specifies the customer's total charges for long-distance calls that exceed those outlined in their plan, as of the end of the aforementioned quarter.
Total Revenue	It specifies the company's total revenue from this customer, calculated as of the end of the aforementioned quarter (Total Charges - Total Refunds + Total Extra Data Charges + Total

	Long-Distance Charges)
Customer Status	Indicates the status of the customer at the end of the quarter: Churned, Stayed, or Joined.
Churn Category	A high-level category for the reason a customer churns, asked when they leave the company: Attitude, Competitor, Dissatisfaction, Other, Price (directly tied to the Churn Reason).
Churn Reason	The specific reason a customer gives for leaving the company, asked at the time of departure (directly linked to the Churn Category).
Zip Code	The postal code of the customer's primary residence.
Population	An up-to-date population estimate for the entire Zip Code area.

Figure 11: Features along with their description (self-generated)

These attributes are divided into 26 categorical and 14 numerical columns, capturing a wide range of customer information, including demographics, subscription details, financial data, and customer behavior indicators.

3.1.1.1 Data Types and Dataset Attributes

The dataset consists of various data types, further classified as:

- **Numerical Columns:** 9 features are of type float64, and 6 features are of type int64. These typically include features like monthly charges, tenure, and total revenue.
- **Categorical Columns:** 23 features are of type object, including variables such as customer status, internet type, and contract type.

The dataset covers several aspects of customer information:

- **Demographics:** Attributes such as age, gender, and marital status.
- **Subscription Details:** Information regarding phone services, internet type, and contract length.
- **Financial Data:** Includes features like monthly charges, total revenue, and payment methods.
- **Customer Behaviour Indicators:** Number of dependents, churn reason, churn status, and other related behaviour patterns.

This diverse range of data types contributes to a comprehensive understanding of customer behaviour, making it possible to analyse both quantitative and qualitative attributes for churn prediction.

3.1.1.2 Target Variable

The **"Customer Status"** column is the target variable in this research and plays a critical role in the churn prediction model. It indicates the status of the customers and serves as the classification label for predicting customer behaviour. The possible values of the "Customer Status" variable are:

- **Stayed:** Customers who have continued their subscription and remain with the company.
- **Joined:** New customers who have recently subscribed to the company's services.

- **Churn:** Customers who have terminated their subscriptions and no longer use the services.

3.2 Data Pre-Processing

Malley et al. (2016) advocated that data preprocessing is a crucial step in transforming raw data into a format suitable for analysis. It consists of a series of operations aimed at improving data quality.

For this research, the preprocessing phase was focused on the following key steps: importing essential libraries, handling missing values, encoding categorical variables, and scaling numerical features.

3.2.1 Importing Libraries

To perform data preprocessing and build machine learning models for customer churn prediction, several essential libraries were imported as illustrated in Figure 12.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import classification_report
from sklearn.metrics import classification_report, accuracy_score
from sklearn.cluster import KMeans
```

Figure 12 showing essential Python libraries used for building the customer churn prediction model.

The libraries used in this research; Pandas, Matplotlib, Seaborn, and scikit-learn, form an essential toolkit that navigated the study. ‘Pandas’, an efficient library, make it easy to manipulate and analyze datasets, allowing for seamless data handling. ‘Matplotlib’ and Seaborn complement this by offering powerful visualization tools that bring data to life, helping to reveal intricate patterns and relationships. Meanwhile, ‘Scikit-learn’ provides a rich collection of functionalities for building machine learning models, including various classification algorithms and performance evaluation metrics. Together, these libraries simplify the entire process of importing datasets, visualizing data, constructing predictive

models, and assessing their effectiveness, ultimately creating a solid foundation for an effective customer churn prediction model.

3.2.2 Loading the Dataset

The next step involved loading the dataset. As the dataset was in CSV format so to import the dataset in Python, 'Pandas' library was utilized.

```
# Importing Dataset
data = pd.read_csv(r'Data\Telecom+Customer+Churn\telecom_customer_churn.csv')
```

Figure 13: Loading the dataset

Upon importing the dataset, all 7043 rows and 38 columns were loaded. The processing time taken to import the dataset is 0.026823 seconds. The processing time indicates that dataset was efficiently imported.

```
Time taken to import the dataset: 0.026823759078979492 seconds
Number of rows loaded: 7043
Number of columns loaded: 38
```

Figure 14: Dataset Loading Summary

3.2.3 Handling Missing Values

Handling missing values is an essential step in ensuring the quality and completeness of the dataset. As Emmanuel et al. (2021) noted, missing values can occur due to several factors, such as system malfunctions during data collection or human error during preprocessing. Failure to address these missing values can lead to biased or inaccurate conclusions.

In this research, handling missing values involved identifying the affected columns and applying appropriate strategies for numerical and categorical columns to ensure the dataset was ready for further analysis and model building.

i. Identifying Columns with Missing Data

The first step involved identifying which columns in the dataset contained missing values, which was essential in determining which features required attention during preprocessing. The 'data.info ()' function was used to display the non-null count for all the features. Out of 38 total columns, 15 columns were found to have missing values. Figure 15 below shows the output of the 'data.info()' function:

```
Data columns (total 38 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Customer ID                        7043 non-null   object
 1   Gender                             7043 non-null   object
 2   Age                                7043 non-null   int64
 3   Married                            7043 non-null   object
 4   Number of Dependents               7043 non-null   int64
 5   City                               7043 non-null   object
 6   Zip Code                           7043 non-null   int64
 7   Latitude                           7043 non-null   float64
 8   Longitude                          7043 non-null   float64
 9   Number of Referrals                7043 non-null   int64
 10  Tenure in Months                   7043 non-null   int64
 11  Offer                              3166 non-null   object
 12  Phone Service                      7043 non-null   object
 13  Avg Monthly Long Distance Charges  6361 non-null   float64
 14  Multiple Lines                     6361 non-null   object
 15  Internet Service                   7043 non-null   object
 16  Internet Type                      5517 non-null   object
 17  Avg Monthly GB Download            5517 non-null   float64
 18  Online Security                    5517 non-null   object
 19  Online Backup                      5517 non-null   object
 20  Device Protection Plan             5517 non-null   object
 21  Premium Tech Support               5517 non-null   object
 22  Streaming TV                       5517 non-null   object
 23  Streaming Movies                   5517 non-null   object
 24  Streaming Music                    5517 non-null   object
 25  Unlimited Data                     5517 non-null   object
 26  Contract                           7043 non-null   object
 27  Paperless Billing                  7043 non-null   object
 28  Payment Method                     7043 non-null   object
 29  Monthly Charge                     7043 non-null   float64
 30  Total Charges                      7043 non-null   float64
 31  Total Refunds                      7043 non-null   float64
 32  Total Extra Data Charges           7043 non-null   int64
 33  Total Long Distance Charges        7043 non-null   float64
 34  Total Revenue                      7043 non-null   float64
 35  Customer Status                    7043 non-null   object
 36  Churn Category                     1869 non-null   object
 37  Churn Reason                       1869 non-null   object
```

Figure 15: Summary of Non-null Counts in the dataset

```
Columns with Missing Values:
- Offer
- Avg Monthly Long Distance Charges
- Multiple Lines
- Internet Type
- Avg Monthly GB Download
- Online Security
- Online Backup
- Device Protection Plan
- Premium Tech Support
- Streaming TV
- Streaming Movies
- Streaming Music
- Unlimited Data
- Churn Category
- Churn Reason
```

Figure 16:list of all columns with Missing Values

The output shown in Figure 16 lists all the columns with missing values in the dataset, making it easy to pinpoint where to focus during the cleaning process.

ii. Handling Missing Values in Numerical and Categorical Columns

The output shown in Figure 17 identifies 15 columns with missing values, two of which- 'Avg Monthly Long-Distance Charges' and 'Avg Monthly GB Download' are numerical. Based on the nature of data, the missing values were handled by filling them with 0.

```
# Numerical Columns: Filling missing values with '0'
data['Avg Monthly GB Download'].fillna(0, inplace=True)
data['Avg Monthly Long Distance Charges'].fillna(0, inplace=True)
```

Figure 17: representing filling missing values in Numerical Columns

After a thorough analysis, it was identified that the missing values in the 'Avg Monthly GB Download' occurred when the 'Internet Service' feature was marked as 'No', indicating that the customer has not subscribed to the Internet Service. So, it is logical for these charges to be 0, as these customers should not be charged.

Similarly, for 'Avg Monthly Long Distance Charges' missing values occurred when the customer did not subscribe to 'Home Service' feature and it was set to 'No'. So as these customers have no internet usage, the average data usage was set to 0.

This approach ensures that the numerical columns are now complete without significantly altering their original distribution.

For categorical columns, missing values were addressed after analyzing the data patterns in the dataset. The technique varied based on the specific characteristics of each column.

The first categorical column with missing values, ‘Offer’, did not originally contain missing values. Instead, it had the value ‘None’ entered, which Python interprets as a NaN (missing value). So, to fix this, the ‘None’ entries were replaced with ‘No offer’, to accurately reflect that ‘No offer’ was provided, as shown below in Figure 18.

```
data['Offer'] = data['Offer'].replace({'None': 'No offer'})
data['Offer'] = data['Offer'].fillna('No offer')
```

Figure 18: Replacing 'None' with 'No Offer' in the 'Offer' Column

The next categorical column addressed was ‘Multiple Lines.’ The missing values in this column were filled with ‘No,’ as customers who do not subscribe to phone service logically cannot have multiple lines.

Similarly, missing values in all internet-related categorical columns were handled based on the customer’s subscription status. The columns affected were: ‘Internet Type,’ ‘Online Security,’ ‘Online Backup,’ ‘Device Protection Plan,’ ‘Premium Tech Support,’ ‘Streaming TV,’ ‘Streaming Movies,’ ‘Streaming Music,’ and ‘Unlimited Data.’

```
# List of categorical columns related to internet service
internet_service_categorical = ['Internet Type', 'Online Security', 'Online Backup', 'Device Protection Plan', 'Premium Tech Support',
'Streaming TV', 'Streaming Movies', 'Streaming Music', 'Unlimited Data']

# Replace missing values with 'No Internet' for customers without internet
data[internet_service_categorical] = data[internet_service_categorical].fillna('No Internet')
```

Figure 19: Handling Missing Values in Internet-Related Categorical Columns

For all these columns, missing values were replaced with ‘No Internet’ to accurately reflect that these services are not applicable to customers without an internet subscription.

In addition to the columns discussed above, ‘Churn Category’ and ‘Churn Reason’ also contain missing values. However, these were not addressed, as they are dependent on the target variable (churn) and are excluded from the analysis. Since they do not contribute to the prediction of churn or the model-building process, they were not included in the imputation process.

iii. Verifying the Handling of Missing Values

Once missing values were filled in both numerical and categorical columns, it was crucial to confirm that no missing data remained. This was done by running the ‘info ()’ function, which provides an overview of the dataset, as shown in Figure 20.

0	Customer ID	7043 non-null	object
1	Gender	7043 non-null	object
2	Age	7043 non-null	int64
3	Married	7043 non-null	object
4	Number of Dependents	7043 non-null	int64
5	City	7043 non-null	object
6	Zip Code	7043 non-null	int64
7	Latitude	7043 non-null	float64
8	Longitude	7043 non-null	float64
9	Number of Referrals	7043 non-null	int64
10	Tenure in Months	7043 non-null	int64
11	Offer	7043 non-null	object
12	Phone Service	7043 non-null	object
13	Avg Monthly Long Distance Charges	7043 non-null	float64
14	Multiple Lines	7043 non-null	object
15	Internet Service	7043 non-null	object
16	Internet Type	7043 non-null	object
17	Avg Monthly GB Download	7043 non-null	float64
18	Online Security	7043 non-null	object
19	Online Backup	7043 non-null	object
20	Device Protection Plan	7043 non-null	object
21	Premium Tech Support	7043 non-null	object
22	Streaming TV	7043 non-null	object
23	Streaming Movies	7043 non-null	object
24	Streaming Music	7043 non-null	object
25	Unlimited Data	7043 non-null	object
26	Contract	7043 non-null	object
27	Paperless Billing	7043 non-null	object
28	Payment Method	7043 non-null	object
29	Monthly Charge	7043 non-null	float64
30	Total Charges	7043 non-null	float64
31	Total Refunds	7043 non-null	float64
32	Total Extra Data Charges	7043 non-null	int64
33	Total Long Distance Charges	7043 non-null	float64
34	Total Revenue	7043 non-null	float64

Figure 20 showing dataset overview after handling missing data

3.2.4 Standardizing Target Variable

Once the missing values were handled, the next step involved addressing inconsistencies in the target variable, Customer Status. In the dataset, customers who had 'Joined' were considered to have 'Stayed', as they had not churned yet. Therefore, the value 'Joined' was replaced with 'Stayed' to reflect the accurate status of these customers. This was implemented using the following code:

```
# Replace 'Joined' with 'Stayed' in the 'Customer Status' column
data['Customer Status'] = data['Customer Status'].replace('Joined', 'Stayed')
```

Figure 21: Code snippet showing the replacement of 'Joined' with 'Stayed' in the 'Customer Status' column for consistency in target variable

This step was important to maintain the accuracy of the target variable, ensuring that 'Customer Status' reflected only two possible states: 'Churned' or 'Stayed.' By doing this, the dataset became suitable for binary classification.

3.2.5 Encoding Categorical Variables

Another essential step in data preprocessing was to encode the categorical variables. Since machine learning models require numerical inputs, categorical features must be transformed into a format suitable for the algorithm. One of the most common approaches to achieve this is one-hot encoding. This technique works by assigning a unique numerical value to each category and then converting it into its binary representation (Seger, 2018).

In this research, one-hot encoding was applied to the categorical columns using the 'get_dummies()' function from the pandas library. This function automatically creates binary columns for each category, representing the presence or absence of that category in the data. Figure 22 demonstrates how one-hot encoding was implemented:

```
# One-hot encoding categorical columns
encoded_data = pd.get_dummies(data, columns=['Gender', 'Married', 'Offer', 'Phone Service', 'Multiple Lines',
'Internet Service', 'Internet Type', 'Online Security', 'Online Backup', 'Device Protection Plan',
'Premium Tech Support', 'Streaming TV', 'Streaming Movies', 'Streaming Music',
'Unlimited Data', 'Contract', 'Paperless Billing', 'Payment Method', 'Churn Category'], dtype=int)
```

Figure 22 illustrating One-hot encoding Categorical Columns

Additionally, the target variable 'Customer Status' was also mapped to numerical values, where 'Churned' was assigned a value of 0, and 'Stayed' was assigned a value of 1. This was done to convert the target variable into a binary classification problem, as demonstrated in the code below:

```
# Mapping Target Variable to numerical values
encoded_data['Customer Status'] = encoded_data['Customer Status'].replace({'Churned': 0, 'Stayed': 1})
```

Figure 23 demonstrating mapping of Target Variable to Numerical Values

The **pie chart** below illustrates the proportion of each category within the "Customer Status" variable. As shown, **67%** of the customers have stayed with the company, while **26.5%** have

churned, and **6.4%** have recently joined. This breakdown provides valuable insight into the distribution of customer behaviors, highlighting that the majority are retained customers.

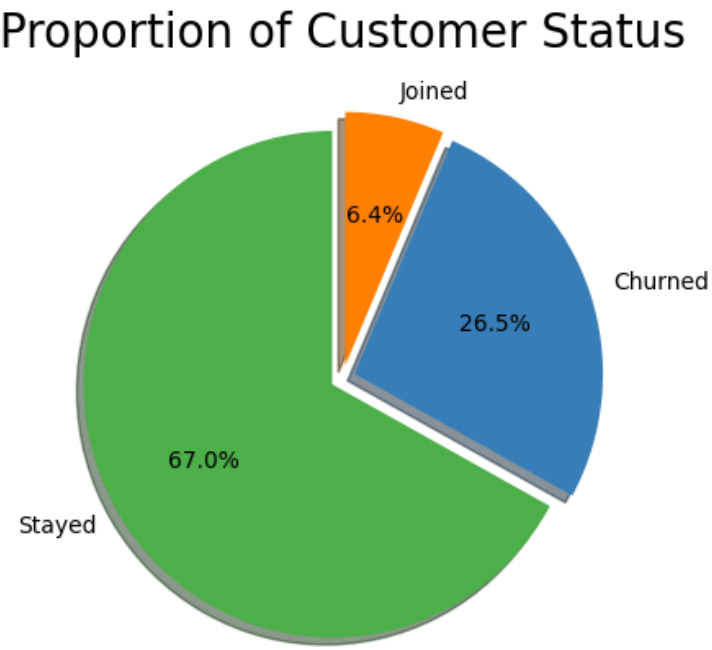


Figure 24 showing Percentages of Customer status

By transforming the categorical features into the numerical ones, the dataset was effectively prepared for use in machine learning models.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) plays a crucial role in understanding the characteristics of a dataset. Its primary objective is to explore the data's distribution, identify outliers, and detect any irregularities that may influence subsequent analysis. As emphasized by (Komorowski et al., 2016), EDA is an important step following data collection and preprocessing, allowing the data to be visualized and examined without assumptions. This approach helps assess the quality of the dataset and informs decisions for model building. Originally introduced by Tukey in 1977 (Páez & Boisjoly, 2022), EDA has since become the standard method for preliminary data analysis. By visually inspecting and exploring the dataset, researchers can uncover key patterns and relationships which then guide the next steps in the research process.

In this research, the goal of this step is to explore the dataset thoroughly, ensuring that it is ready for model building. Several key techniques of EDA are applied to gain insights into the

dataset, as follows:

3.3.1 Descriptive Statistics and Visualizations

The first step in the EDA process for this research involved generating descriptive statistics for both numerical and categorical columns. Descriptive statistics provide an overview of the dataset by summarizing key measures such as mean, median, standard deviation, and counts for each variable. These statistics are typically obtained using the 'describe ()' function for numerical columns and the 'value_counts()' function for categorical columns in pandas.

	Age	Number of Dependents	Zip Code	Latitude	...	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue
count	7043.000000	7043.000000	7043.000000	7043.000000	...	7043.000000	7043.000000	7043.000000	7043.000000
mean	46.509726	0.468692	93486.070567	36.197455	...	1.962182	6.860713	749.099262	3034.379056
std	16.750352	0.962802	1856.767505	2.468929	...	7.902614	25.104978	846.660055	2865.204542
min	19.000000	0.000000	90001.000000	32.555828	...	0.000000	0.000000	0.000000	21.360000
25%	32.000000	0.000000	92101.000000	33.990646	...	0.000000	0.000000	70.545000	605.610000
50%	46.000000	0.000000	93518.000000	36.205465	...	0.000000	0.000000	401.440000	2108.640000
75%	60.000000	0.000000	95329.000000	38.161321	...	0.000000	0.000000	1191.100000	4801.145000
max	80.000000	9.000000	96150.000000	41.962127	...	49.790000	150.000000	3564.720000	11979.340000

[8 rows x 15 columns]

Figure 25 showing Descriptive Statistics for Key Numerical Features

Key insights from the descriptive statistics as shown in Figure 25 highlight that the average **Age** is 46.5 years, with most customers having few or no dependents. **Tenure**, an important predictor of churn, averages 32 months, indicating a diverse range of customer loyalty. **Monthly charges** average $63.60, with significant variation, reflecting different service levels or plans. **Total charges** range widely, from $18.80 to $8,684.80, suggesting different levels of service engagement. **Long-distance usage** also varies, with some customers making no calls, while others are heavy users, which may influence churn risk. Finally, **total revenue** distribution shows that some high-value customers contribute disproportionately, making their retention critical.

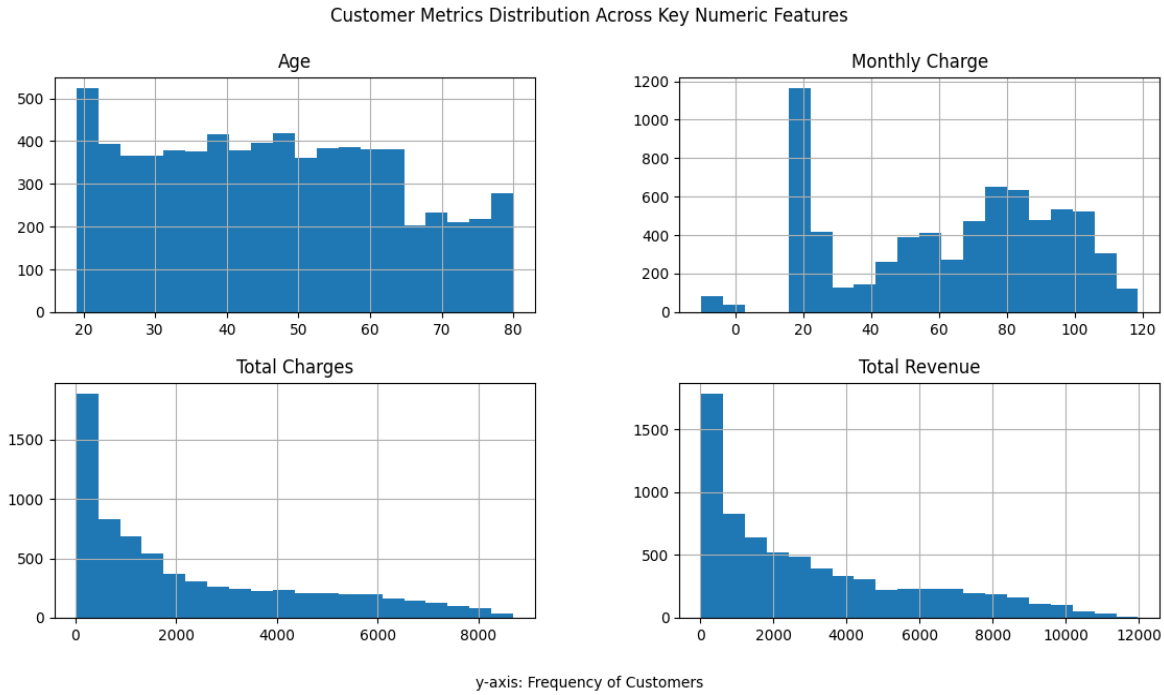


Figure 26: Histogram visualizing some key numeric features

Figure 26 consists of histograms illustrating the distribution of customer metrics across some key numerical features. The x-axis represents the value for each metric, while the y-axis shows the frequency of customers. While the 'Age' column shows relatively uniform distribution, the rest of the columns exhibit skewed distributions, indicating that most of the customers have lower contributions towards charges and revenue.

Similarly, the key insights from the descriptive statistics for categorical columns highlight several factors that may influence churn as shown in Figure 27. **Offer B** is the most popular, with 4,701 customers, while other offers like **Offer E** and **Offer D** have significantly fewer customers. **Phone Service** is used by the majority (6,361), while a smaller group (682) does not use this service, which might indicate lower engagement. The use of **Internet Services** is also prevalent, with 5,517 customers opting for it, which may play a role in satisfaction and churn.

In terms of additional services, many customers have opted out of services like **Online Security**, **Backup**, and **Device Protection**, which may affect churn rates if they feel less protected or serviced. Most customers (3,610) are on **Month-to-Month** contracts, a key churn indicator, as these customers tend to have more flexibility to switch providers. Additionally, 6,271 customers have **Unlimited Data**, suggesting that those without this feature may be more prone to leave if they face data overages or constraints.

In **Churn Category, Competitor offers** is the top reason for churn (6,015), followed by

Dissatisfaction and **Price** concerns. This suggests that competitor pricing and service quality play significant roles in customer retention, which companies must address to reduce churn risk.

KEY CATEGORICAL COLUMNS VALUE COUNT REPORT

Offer	Count
Offer B	4701
Offer E	805
Offer D	602
Offer A	520
Offer C	415

Contract	Count
Month-to-Month	3610
Two Year	1883
One Year	1550

Internet Type	Count
Fiber Optic	4561
DSL	1652
Cable	830

Payment Method	Count
Bank Withdrawal	3909
Credit Card	2749
Mailed Check	385

Phone Service	Count
Yes	6361
No	682

Online Security	Count
No	5024
Yes	2019

Unlimited Data	Count
Yes	6271
No	772

Churn Category	Count
Competitor	6015
Dissatisfaction	321
Attitude	314
Price	211
Other	182

Internet Service	Count
Yes	5517
No	1526

Online Backup	Count
No	4614
Yes	2429

Device Protection Plan	Count
No	4621
Yes	2422

Figure 27 showing the Report of Key Categorical Columns Value Counts

One of the most essential analysis that was done as a part of EDA in this research was the visualization of key categorical data by using ‘groupby()’ and ‘size()’ function in Pandas. These functions groups the data by both feature and target variable and counts the number of customers in each group respectively.

Figure 28 and 29 shows the use of ‘groupby()’ function to group data by both Contract and Customer Status, showing the counts of customers for each of these combinations.

```
# Grouping the data by 'Contract' and 'Customer Status' and counting occurrences
grouped_data = data.groupby(['Contract', 'Customer Status']).size().reset_index(name='Count')

# Display the grouped data (optional to view how it looks)
print(grouped_data)

# Creating the bar plot from the grouped data
sns.barplot(data=grouped_data, x='Contract', y='Count', hue='Customer Status', palette='Spectral')
plt.title('Status Counts by Contract', fontsize=24)
plt.xlabel('Contract', fontsize=18)
plt.ylabel('Count', fontsize=18)
plt.xticks(fontsize=10)
plt.yticks(fontsize=10)
plt.legend(title='Status', fontsize=12)
plt.show()
```

Figure 28: Code to generate a bar plot of customer status counts using grouped data

	Contract	Customer Status	Count
0	Month-to-Month	Churned	1655
1	Month-to-Month	Stayed	1955
2	One Year	Churned	166
3	One Year	Stayed	1384
4	Two Year	Churned	48
5	Two Year	Stayed	1835

Figure 29: Result of Customer Contract Status Count Breakdown: Churned vs. Stayed

The bar chart in Figure 29 provides a comparative illustration of customer churn and retention across different contract types (one-year, month to month, two year). The chart reveals that the customers on month-to-month contracts have the highest number of churn rate, indicating that short-term contracts are more prone to churn. In contract, customers with long-term contracts exhibits significantly lower churn rates.

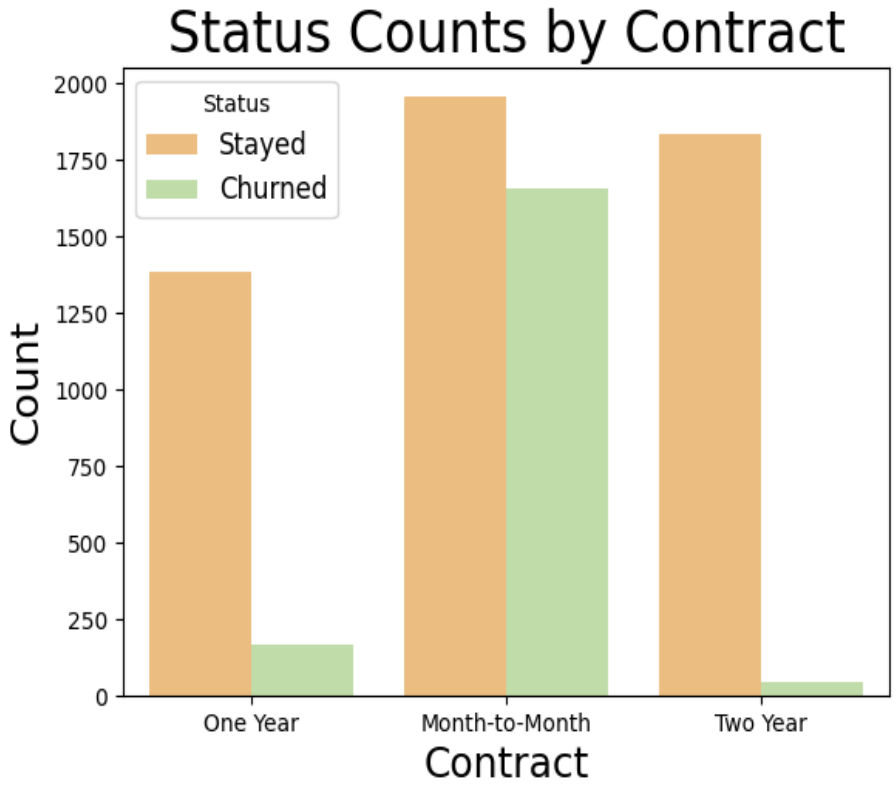


Figure 30: Status count by Contract

Similar analysis was done for two other key features in this dataset i.e.: Phone Service, and Internet Service. For **Phone Service,** the churn rate is slightly higher for customers with a subscription (26.71%) compared to those without (24.93%). For **Internet Service,** the churn rate is much higher for customers with a subscription (31.83%) compared to those without (7.40%).

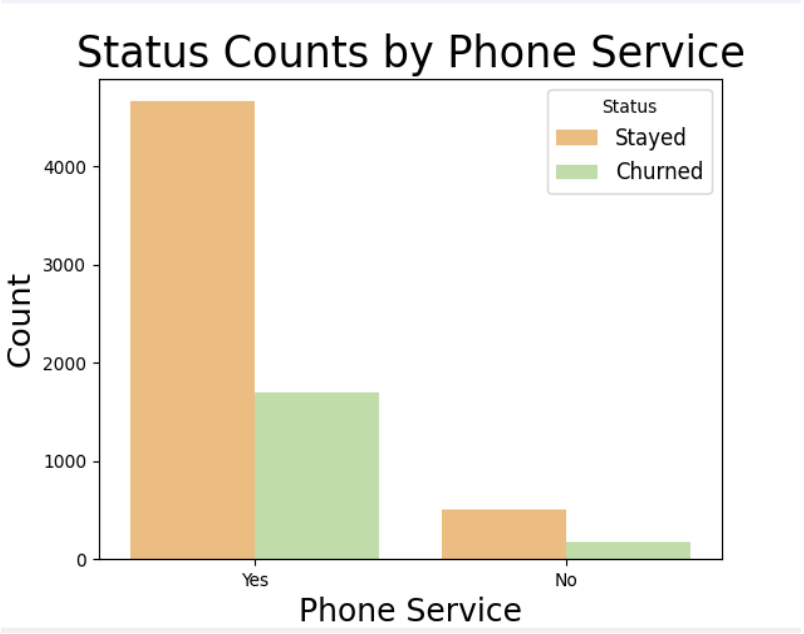


Figure 31: Status count by Phone Service Subscription

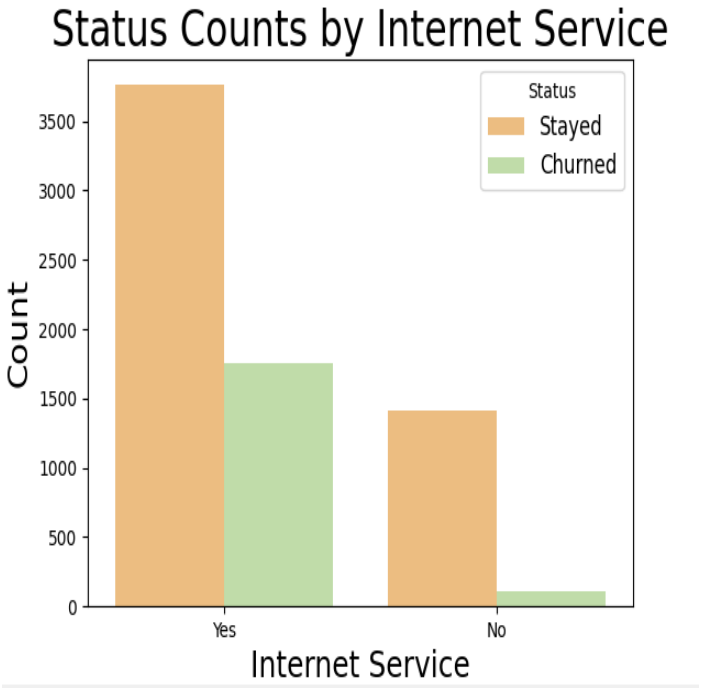


Figure 32: Status Count by Internet Service Subscription

	Phone Service	Customer Status	Count
0	No	Churned	170
1	No	Stayed	512
2	Yes	Churned	1699
3	Yes	Stayed	4662

	Internet Service	Customer Status	Count
0	No	Churned	113
1	No	Stayed	1413
2	Yes	Churned	1756
3	Yes	Stayed	3761

Figure 33:Result of Customer Phone/Internet Status Count Breakdown: Churned vs. Stayed

These visualisations shows that while there is relatively a huge ratio of customers who have phone or internet subscriptions stays to those who do not, but the ratio of churned to stayed is higher for those with service.

The analysis done above is valuable in helping business create targeted marketing retention strategies. For the above key features, understanding which customer segments (e.g, contract type, service subscriptions) are more likely to churn, the company can proactively focus on these groups. For instance, based on our analysis and results this telecommunication company should consider promoting long/yearly contracts and provide specialized offers and incentives to customers who avails phone and internet services to improve retention rates.

Furthermore, this descriptive analysis plays an important role to identify the key features that are likely to influence churn. By examining trends in both numerical and categorical data, this research narrows down the most relevant features to build a robust machine learning churn prediction model, improving the overall effectiveness of the analysis.

3.3.2 Analysing Correlations between Variables

In addition to descriptive statistics and visualisations, EDA also involved analysing the correlation between various variables. The correlation coefficient, introduced by Karl Pearson, is a widely used statistical method to measure the strength and direction of the association between two variables. Janse et al. (2021) explained correlation by using an example where two variables ‘X’ and ‘Y’, are plotted against each other on a scatter plot. The correlation coefficient, which ranges from -1 to 1, indicates how well a straight line fits the data. A value of -1 indicates a perfect negative correlation, 1 represents a perfect positive correlation, whereas 0 means no linear relationship.

As a part of this research, correlation analysis was done to uncover relationships between features and the target variable, Customer Status. The correlation values (Figure 34) and the heatmap (Figure 35) as shown below, gives a clear representation of the relationship across various features.

```
Correlation values:
Customer Status                          1.000000
Tenure in Months                         0.352861
Contract_Two Year                        0.328195
Number of Referrals                      0.286540
Internet Service_No                      0.227890
Internet Type_No Internet                0.227890
Online Security_No Internet              0.227890
Online Backup_No Internet                0.227890
Streaming Music_No Internet              0.227890
Device Protection Plan_No Internet       0.227890
Streaming TV_No Internet                 0.227890
Unlimited Data_No Internet               0.227890
Streaming Movies_No Internet             0.227890
Premium Tech Support_No Internet         0.227890
Total Long Distance Charges              0.223756
Total Revenue                            0.223003
Number of Dependents                     0.218780
Payment Method_Credit Card               0.218528
Total Charges                            0.198546
Paperless Billing_No                     0.191825
Contract_One Year                        0.190418
Online Security_Yes                      0.171226
Premium Tech Support_Yes                 0.164674
Married_Yes                              0.150448
Offer_Offer A                            0.126654
Offer_Offer B                            0.117723
Internet Type_DSL                        0.099716
Online Backup_Yes                        0.082255
Device Protection Plan_Yes               0.066160
Multiple Lines_No                        0.040102
Total Refunds                            0.033709
Offer_Offer C                            0.020660
Phone Service No                         0.011942
Offer_No offer                          -0.014326
Multiple Lines_Yes                      -0.040102
Streaming Music_Yes                     -0.045587
Avg Monthly GB Download                 -0.048868
Unlimited Data_No                       -0.050577
Payment Method_Mailed Check             -0.056348
Streaming Movies_Yes                    -0.061382
Streaming TV_Yes                        -0.063228
Age                                     -0.115760
Streaming TV_No                         -0.128916
Streaming Movies_No                     -0.130845
Streaming Music_No                      -0.145622
Married_No                              -0.150448
Unlimited Data_Yes                      -0.166545
Monthly Charge                          -0.187833
Payment Method_Bank Withdrawal          -0.188733
Paperless Billing_Yes                   -0.191825
Offer_Offer E                           -0.214648
Internet Service_Yes                    -0.227890
Device Protection Plan_No               -0.252481
Online Backup_No                        -0.268005
Internet Type_Fiber Optic               -0.279623
Premium Tech Support_No                 -0.337281
Online Security_No                      -0.342637
Contract Month-to-Month                 -0.448427
```

Figure 34: Correlation Matrix for Key Correlated features

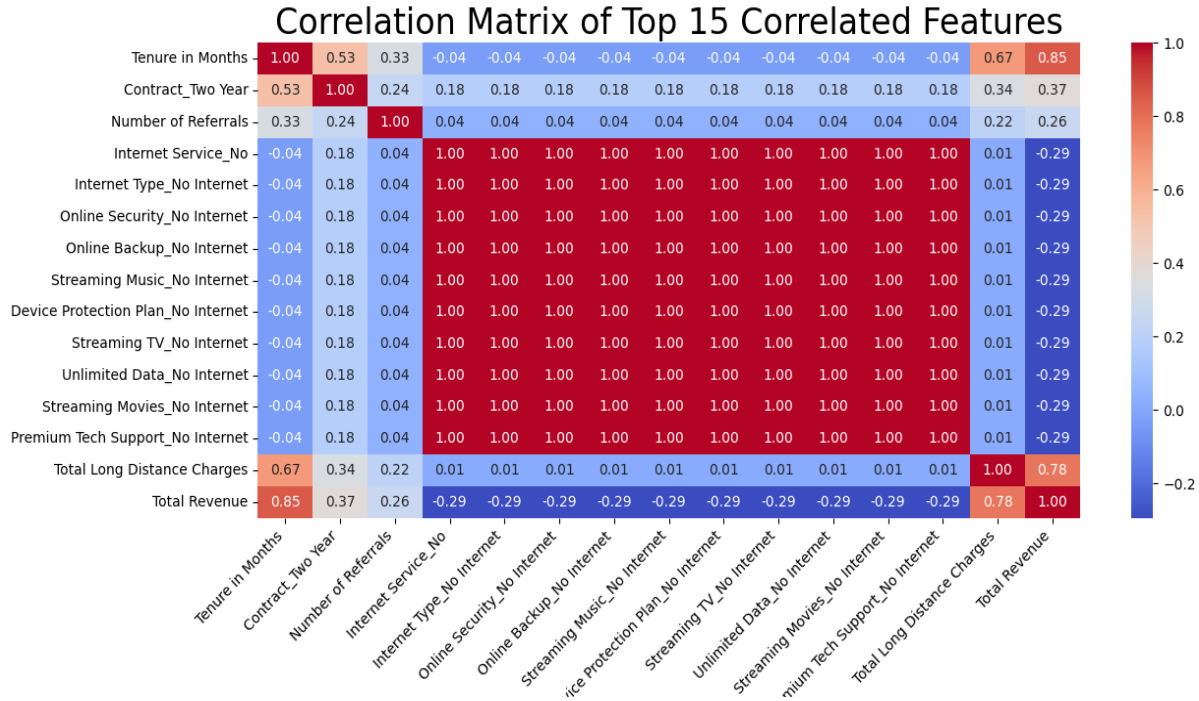


Figure 35: Heatmap of Correlations Between Key Features

The heatmap in Figure 35 represents this relationship using color gradients to indicate the strength of correlations. Red represents stronger correlations while darker shades of blues represent weaker relationships.

Some of the key insights derived from Correlation Matrix and Heatmap includes:

1. **Contract Type and Tenure**: As evident from both Figures 34 and 35, **Contract Month-to-Month** (has a strong negative correlation, i,e -0.44) and **Tenure in Months**(having a moderate positive correlation, i,e 0.35) are two of the most critical features when predicting customer churn. The customers availing month-to-month contracts are more likely to leave, while those with longer tenures are predicted to stay.
2. **Online Services and Internet Plans:** The matrix shows that features like **Online Security_No** (-0.34), **Premium Tech Support_No** (-0.33), and **Device Protection Plan_No** (-0.25) have relatively strong negative correlations with customer status. This means that customers who have not subscribed to these additional online services (security, tech support, or device protection) are more likely to churn.
   Furthermore, features related to Internet service (Streaming TV, Online Backup) exhibits moderate positive correlations, implying that customers who don't avail internet services are less likely to churn, possibly because they aren't relying on the service as much.

3. **Number of Referrals:** This variable also has a positive correlation of 0.28 with the target variable, indicating that customers who refer others tends to be more loyal and are less likely to churn.

By analyzing these insights, it becomes clear that focusing on long-term contracts, encouraging service upgrades, and promoting a strong referral program can significantly reduce churn. Additionally, placing a targeted emphasis on this high-risk area can enable the business to implement more effective retention strategies, ultimately improving customer loyalty.

3.4 Feature Selection

After analysing the correlation between variables, the next step was to select important features from the dataset. Pudjihartono et al. (2022) highlighted that feature selection is widely used to address the issue of excessive and irrelevant features. It is a critical step in the process, as effective feature selection can improve learning efficiency, model performance and accuracy by reducing dimensionality.

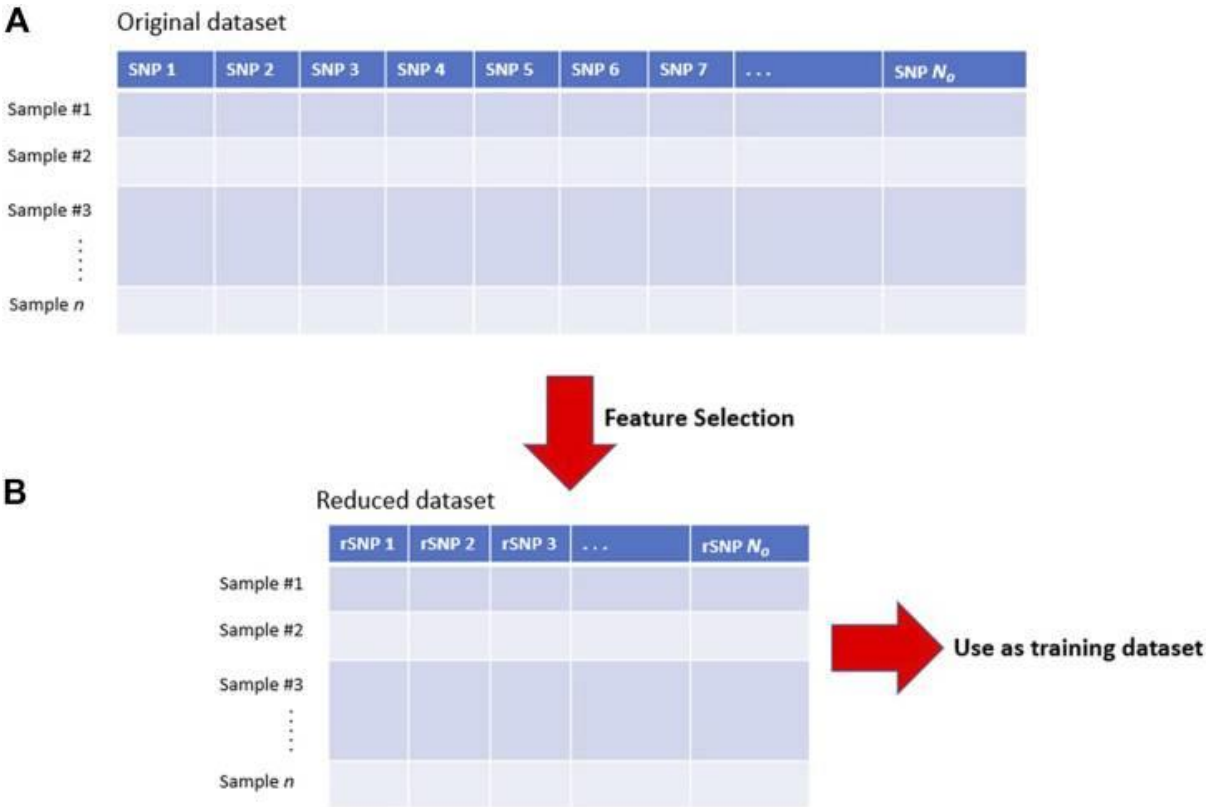


Figure 36: Feature Selection Process (Pudjihartono et al., 2022)

Figure 36 illustrates the process of feature selection in detail. In the original dataset (A) all available features are present, including some that may be irrelevant to the machine learning

model. After feature selection, the dimensionality is reduced, leaving the only most relevant features in the reduced dataset (B). This dataset is then used for training the algorithm.

3.4.1 Feature Selection with Correlation Analysis

For this research, the feature selection was initially performed by examining the correlation between each independent variable and the target variable, which indicates whether a customer has churned or stayed.

```
# Correlation matrix
correlation_matrix = corr_df.corr()
Customer_Status_corr = correlation_matrix['Customer Status']

high_correlation = Customer_Status_corr[Customer_Status_corr.abs() > 0.01].sort_values(ascending=False)
print(" ")
print("Correlation values:")
print(high_correlation)
```

Figure 37: Code to compute correlations with 'Customer Status' above 0.01

Using the code as shown in Figure 37, correlation values were sorted based on their absolute strength. A threshold of 0.15 was applied, as typically this level represents a moderate relationship with the target variable. Features meeting or exceeding this threshold, were taken into consideration for further analysis. Figure 38 illustrates the 15 top correlated features that initially chosen for model training, with detailed model building steps provided in section 2.3.5, as they showed a reasonably strong relationship with customer churn.

```
top_corr_features = ['Tenure in Months', 'Contract_Two Year', 'Number of Referrals', 'Internet Service_No', 'Internet Type_No Internet',
'Online Security_No Internet','Online Backup_No Internet', 'Streaming Music_No Internet',
'Device Protection Plan_No Internet','Streaming TV_No Internet', 'Unlimited Data_No Internet',
'Streaming Movies_No Internet','Premium Tech Support_No Internet','Total Long Distance Charges', 'Total Revenue']
```

Figure 38: Top 15 Correlated Features selected for model training

Using these features the logistic regression model was trained, resulting in an accuracy of 78%. The Classification report as shown in Figure 39 highlights that the model performed well in predicting customers who stayed (Class 1), achieving a precision of 0.84 and a recall of 0.86. In contrast, the precision for predicting customers who churned (Class 0) was lower, at 0.59, with a recall of 0.55, indicating that while the model excelled in detecting the likelihood of customers staying in the company, it struggled to effectively detect churners.

```
Logistic Regression Accuracy with Top 15 Correlated Features: 78.00%

Classification Report with Top 15 Correlate Features:
              precision    recall  f1-score   support

           0       0.59      0.55      0.57       373
           1       0.84      0.86      0.85      1036

    accuracy                           0.78      1409
   macro avg       0.72      0.71      0.71      1409
weighted avg       0.78      0.78      0.78      1409
```

Figure 39: Logistic Regression Accuracy using Top 15 Correlated Features

Since the model's performance was suboptimal with the initial set of features, the feature selection process was refined. The updated feature set as shown in Figure 40, focused on variables with high predictive power based on their correlation with the target variable 'Customer Status'.

```
top_corr_features = ['Tenure in Months', 'Contract_Two Year', 'Number of Referrals','Number of Dependents',
'Payment Method_Credit Card', 'Total Charges', 'Paperless Billing_No', 'Contract_One Year',
'Online Security_Yes', 'Premium Tech Support_Yes', 'Married_Yes', 'Offer_Offer A',
'Total Long Distance Charges', 'Total Revenue']
```

Figure 40: Updated Top Correlated Feature set for Model training

With this revised feature set, a significant increase in model accuracy was observed, reaching 82.61%, showcasing a stronger ability to predict customer churn when compared to the previous set of features.

```
Logistic Regression Accuracy with Top 15 Correlated Features: 82.61%

Classification Report with Top 15 Correlate Features:
              precision    recall  f1-score   support

           0       0.67      0.68      0.67       373
           1       0.88      0.88      0.88      1036

    accuracy                           0.83      1409
   macro avg       0.78      0.78      0.78      1409
weighted avg       0.83      0.83      0.83      1409
```

Figure 41: Logistic Regression Accuracy using refined Top 15 Correlated Features

The improvement in model's accuracy can be due to several reasons, primarily due to the addition of features with stronger, more direct relationships to customer churn and the removal of irrelevant and redundant variables. Features like "No Internet" or "No Streaming Services" were less effective, as they reflected a lack of service rather than focusing on the customer behavior, which is critical for predicting churn. And the inclusion of features like 'Number of Dependents', 'Total Charges' provided more insights into customer demographics and financial behavior which are more effective when understanding churn

likelihood.

3.4.2 Feature Selection with Random Forest

To refine feature selection, Random Forest algorithm was employed to calculate feature importance. This method was utilized to capture non-linear interactions between the features and the target variable.

Feature importance in the Random Forest algorithm is measured by how much each feature reduces impurity during splits, using a metric called Gini importance. Features with higher Gini importance have a greater impact on the model's predictions (Prasanna Ghattikar, 2023).

As shown in Figure 42, the code maps the importance scores to the respective feature names and ranks them based on their contribution to the model. The top 15 features were identified through this process and were then selected for model training. Key variables, as shown in Figure 43, like 'Contract_Month-to-Month', 'Monthy Charge' and 'Tenure in Months', were selected as they not only represent direct customer behaviors and interactions with the service but also provides insight into long-term engagement, financial aspects and service usage patterns.

```
# Feature importance with RandomForest
importances = rf.feature_importances_
# Mapping importances to feature names
feature_importances = pd.Series(importances, index=X.columns)

# Sorting feature importances
feature_importances = feature_importances.sort_values(ascending=False)
print(" ")
print("Feature Importance Values are:")
print(feature_importances)

# Selecting top 15 feature importances
top_15_features = feature_importances.head(15)
print("Top 15 Feature Importance Values are:")
print(top_15_features)

top_feature_importance = ['Contract_Month-to-Month', 'Monthly Charge', 'Tenure in Months', 'Total Revenue', 'Age',
                          'Total Charges', 'Number of Referrals', 'Total Long Distance Charges', 'Zip Code',
                          'Avg Monthly Long Distance Charges', 'Avg Monthly GB Download', 'Contract_Two Year',
                          'Premium Tech Support_No', 'Internet Type_Fiber Optic', 'Number of Dependents']
```

Figure 42: Calculating feature importance through Random Forest

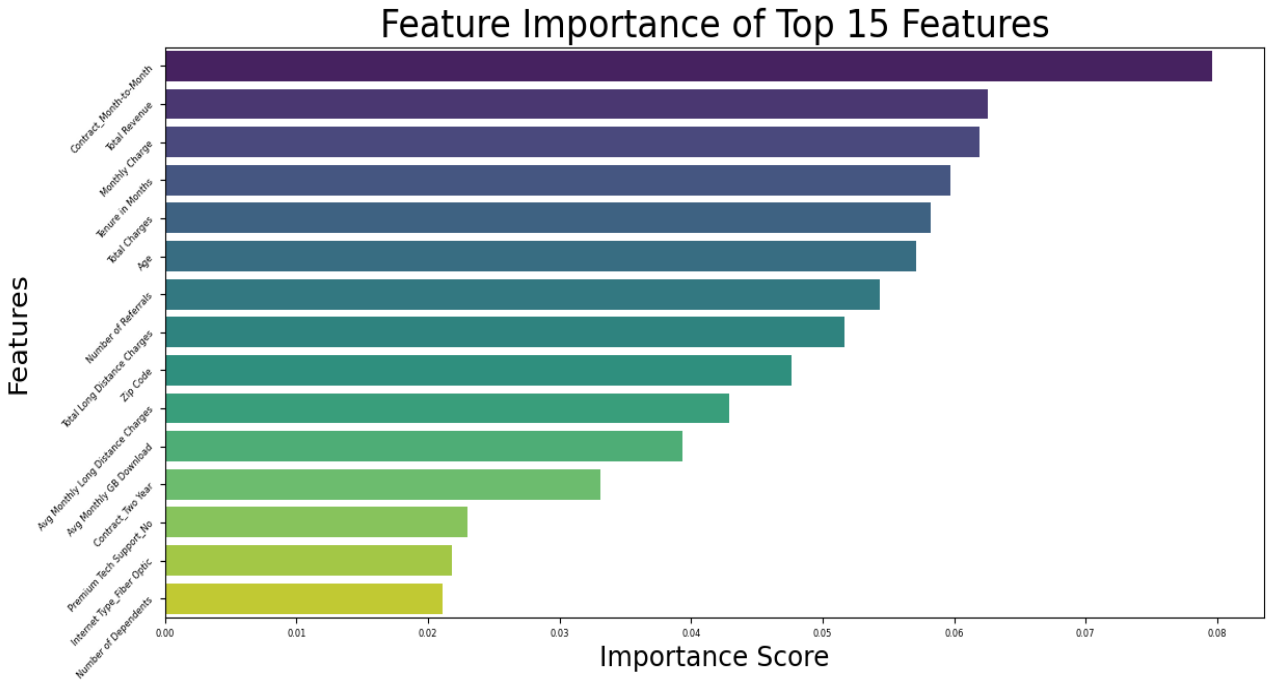


Figure 43: Feature Importance Bar Chart

Using this refined feature set, a logistic regression model was trained, yielding an accuracy of 81.55%. This classification report, as seen in Figure 44, shows that the model performed well overall, particularly in identifying customers likely to stay, with a balanced average accuracy of 82%.

```
Logistic Regression Accuracy with Top 15 Important Features: 81.55%

Classification Report with Top 15 Important Features:
              precision    recall  f1-score   support

           0       0.65      0.66      0.65       373
           1       0.88      0.87      0.87      1036

    accuracy                           0.82      1409
   macro avg       0.76      0.77      0.76      1409
weighted avg       0.82      0.82      0.82      1409
```

Figure 44: Logistic Regression Accuracy using Top 15 Feature Importance variables

Both feature selection methods explored in this study, Correlation analysis and Random Forest feature importance, resulted in a refined set of features that only improved model accuracy but also highlighted valuable aspects of customer behaviour most indicative of churn. This lays a strong foundation for further analysis and modelling.

3.4.3 Utilizing Feature Selection for Customer Segmentation Analysis

The feature selection process was carried out that not only identified key variables but also created a way to work on targeted customer segmentation. And to further enhance the understanding of customer behaviour, customer segmentation was implemented that classified customers into three distinct groups, Prime Customers, Mid-Tier Customers, and High Chun Risk, based on a set of defined criteria. This classification was derived from a combination of attributes which included the top features selected in the feature selection process including contract type, total revenue, monthly charges, tenure, demographic factors and some behavioural metrics.

```
# Function to perform customer segmentation
def customer_segmentation(row):
    # Prime Customers
    if (
        row['Contract_Month-to-Month'] == 0 and  # Assuming 0 for non-month-to-month customers
        row['Total Revenue'] >= 5000 and
        row['Monthly Charge'] <= 80 and
        row['Tenure in Months'] >= 36 and
        row['Total Charges'] >= 2000 and
        row['Age'] >= 40 and
        row['Number of Referrals'] >= 2 and
        row['Total Long Distance Charges'] <= 100 and
        row['Zip Code'] in range(10000, 20000) and  # Assuming premium locations by Zip Code range
        row['Avg Monthly Long Distance Charges'] <= 15
    ):
        return 'Prime'

    # Mid-Tier Customers (Average attributes, moderate terms)
    elif (
        row['Contract_Month-to-Month'] == 1 and
        3000 <= row['Total Revenue'] < 5000 and
        80 < row['Monthly Charge'] <= 100 and
        24 <= row['Tenure in Months'] < 36 and
```

Figure 45: Code for Customer Segmentation Based on Key Attributes (i)

```
        1000 <= row['Total Charges'] < 2000 and
        30 <= row['Age'] < 40 and
        1 <= row['Number of Referrals'] < 2 and
        100 <= row['Total Long Distance Charges'] <= 200 and
        row['Zip Code'] in range(20000, 30000) and
        15 < row['Avg Monthly Long Distance Charges'] <= 30
    ):
        return 'Mid-Tier'

    # High Default Risk (Remaining customers who don't fit into Prime or Mid-Tier categories)
    return 'High Default Risk'

# Sample DataFrame setup (assuming df_imp_features is your dataset)
# Apply the segmentation function to each row of the DataFrame
encoded_data['Customer_Segment'] = encoded_data.apply(customer_segmentation, axis=1)
```

Figure 46:Code for Customer Segmentation Based on Key Attributes(ii)

As illustrated in Figure 45 and 46, this code segments customers into three groups by classifying customers into three groups based on specific criteria derived from key features identified during the feature selection process. The thresholds for each attribute were set using percentiles from the ‘describe ()’ function in pandas. The customer_segmentation(row) function evaluates each row of customer data from the encoded_data DataFrame, applying conditions to classify customers into one of the three categories.

Prime Customers are high-value, loyal clients with long-term contracts, high revenue, low monthly charges, and a history of multiple referrals. Mid-Tier customers offer moderate value, characterized by shorter tenure, higher charges, and less stable, month-to-month contracts. Lastly, High Churn Risk customers fall outside of the Prime or Mid-Tier criteria, indicating a higher likelihood of churn and potential financial instability.

	Customer_Segment	Contract_Month-to-Month	Total Revenue	Monthly Charge	Tenure in Months	Total Charges	Age	Number of Referrals
0	High Churn Risk	0	974.81	65.60	9	593.30	37	2
1	High Churn Risk	1	610.28	-4.00	9	542.40	46	0
2	High Churn Risk	1	415.45	73.90	4	280.85	50	0
3	High Churn Risk	1	1,599.51	98.00	13	1,237.85	78	1
4	High Churn Risk	1	289.54	83.90	3	267.40	75	3
5	High Churn Risk	1	722.38	69.40	9	571.45	23	0
6	High Churn Risk	0	8,611.41	109.70	71	7,904.25	67	1
7	High Churn Risk	0	6,214.28	84.65	63	5,377.80	52	8
8	High Churn Risk	0	414.06	48.20	7	340.35	68	0
9	High Churn Risk	0	7,807.80	90.45	65	5,957.90	43	3
10	High Churn Risk	0	2,460.55	45.20	54	2,460.55	47	0
11	High Churn Risk	0	9,609.47	116.80	72	8,456.75	25	2
12	High Churn Risk	1	444.75	68.95	5	351.50	58	0
13	High Churn Risk	0	7,423.25	101.30	72	7,261.25	32	0
14	High Churn Risk	0	2,560.10	45.05	56	2,560.10	39	0
15	High Churn Risk	0	8,784.86	95.75	71	6,849.40	58	9
16	High Churn Risk	0	1,993.20	61.25	34	1,993.20	52	0
17	High Churn Risk	0	79.87	72.10	1	72.10	72	0
18	High Churn Risk	0	3,271.65	62.70	45	2,791.50	79	0
19	High Churn Risk	1	25.10	25.10	1	25.10	67	0

Figure 47: Customer Segmentation Output: High Churn Risk Customers with Key Attributes

The output presented in Figure 47 reveals that the High Churn Risk segment, compromises customers who are more likely to leave the service or switch to a different service provider. These customers generally tend to have shorter tenures, higher monthly charges, or are on month-to-month contracts, which offer more flexibility for cancellation. Although their total revenue and tenure may vary, the output highlights that customer behaviour and contract type are crucial factors influencing churn risk.

By identifying Prime, Mid-Tier, and High Churn Risk Customers, companies can use their resources more effectively. This targeted customer segmentation allows them to customize their marketing efforts and services for different customer groups, leading to better retention and satisfaction.

3.4.3.1 Utilizing Selected Features for Enhanced Customer Insights and Retention Strategies

The comprehensive feature selection process has provided a refined set of key variables that significantly influence customer churn. These selected features including Contract Type, Monthly Charges, Tenure, Number of Dependents and Total Charges, provide critical insights into customer behavior and preferences.

These top features will come very useful to the business to streamline its onboarding process by limiting customer feedback to only these essential features. This approach comes in handy as it effectively reduces customer's fatigue in early interactions, leading to better engagement and lower chances of churn. Ultimately, these strategies streamline data collection and build customer loyalty by showing a genuine commitment to understanding and meeting individual needs.

3.5 Model Implementation

For this research, two machine learning algorithms, Logistic Regression and Random Forest, were utilized to predict the customer churn. These models were chosen based on their robustness and simplicity in the classification tasks, more specifically when dealing with both categorical and numerical features, which is critical in the context of customer churn prediction.

3.5.1 Logistic Regression

Logistic regression is a widely used statistical tool particularly used for binary classification problems. Developed by David Cox in 1958, this technique was built to calculate the relation between features and probability of particular outcome (Yang, 2019).

The general equation to calculate Logistic Regression is as follows:

$$p(X) = \frac{e^{(b_0 + b_1^* X)}}{1 + e^{(b_0 + b_1^* X)}}$$

Figure 48: General equation of Logistic Regression Yang (2019)

Here p(x) refers to as predicted output, b_0 is the intercept term, whereas b_1 is the coefficient for the single input value (x).

3.5.1.1 Model Implementation

In our research, this model was chosen due to it effectiveness in binary classification tasks

like predicting customer churn, where the outcome is either churn or stay.

After data-preprocessing, exploratory data analysis and standardizing the features, this model was built through a series of steps. The first step was to define the features (independent variables) and target variable (dependent variable: customer churn status). The dataset that was used to define these features was the one that had undergone one-hot encoding.

```
# Defining features and target variable
X = encoded_data.drop(columns=['Customer ID', 'City', 'Latitude', 'Longitude','Customer Status', 'Churn Reason','Churn Category'])
Y = encoded_data['Customer Status']
```

Figure 49: Defining feature and Target Variables for Logistic Regression Model

Figure 49 shows the step in which the feature set '**X**' is defined by dropping specific columns that are either irrelevant or could negatively impact the model's performance. **Customer Id**, **City**, **Latitude**, and **Longitude** were dropped because these consists of geographical or identifier information that does not directly contribute to predicting the customer churn. **Churn Reason** and **Churn Category** are dependent variables that are only available for customers who have already churned; including them would introduce bias into the model. The **Customer Status** column, being the target variable, was also excluded from the feature set to prevent data leakage. On the other hand, '**Y**' is the dependent variable, representing the target variable '**Customer Status**', which indicate whether the customer has churned or stayed.

The next step included training the model using the 20:80 test-train split technique.

```
# Split the dataset into training and testing sets
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

Figure 50: Splitting the dataset into training and testing sets

Figure 50 shows splitting the dataset using 'train_test_split' function, which randomly divides the data into two parts: training and testing sets. The parameter 'test size' =0.2' shows that 20% of the data will be used for testing while 80% of it will be used for training. The 'random_state=2' parameter is done to ensure reproducibility, ensuring that the split is the same every time the code is run, allowing for consistent results.

```
# Initialize the Logistic Regression model
logreg = LogisticRegression(random_state=42)

# Train the model on the scaled training data
logreg.fit(X_train_scaled, Y_train)

# Make predictions on the test set
Y_pred = logreg.predict(X_test_scaled)
```

Figure 51: Logistic Regression Implementation

Figure 51 shows the model initialization, training and prediction on unseen test data. The logistic regression was first initialized and trained on the scaled training data. Afterwards, the model predicts the customer churn statis on the scaled test data(**X_test_scaled**) and the predictions are stored in **'Y_pred'** variable.

3.5.1.2 Results and Accuracy

After building and training the logistic regression model on the pre-processed data, the model was evaluated using the testing dataset. Key metrics such as accuracy, precision, recall and F1-score were employed to assess performance. To evaluate the accuracy of the model, the 'accuracy_score' function was utilized as shown in Figure 52.

According to Singh (n.d.) accuracy is a crucial metric for evaluating a logistic regression model's performance, as it indicates the number of correct predictions out of the total number of predictions.

```
#Evaluate the model
accuracy = accuracy_score(Y_test, Y_pred)
classification_rep = classification_report(Y_test, Y_pred)

# Print the results
print("Logistic Regression Accuracy: {:.2f}%".format(accuracy * 100))
print("\nClassification Report:\n", classification_rep)
```

Figure 52: Logistic Regression Accuracy and Classification Report with all features

The model achieved an overall accuracy of 83.32%, meaning that approximately 83 out of 100 customers were classified correctly as either 'churn' or 'stay'.

```
Logistic Regression Accuracy: 83.32%

Classification Report:
              precision    recall  f1-score   support

           0       0.69      0.67      0.68       373
           1       0.88      0.89      0.89      1036

    accuracy                           0.83      1409
   macro avg       0.79      0.78      0.78      1409
weighted avg       0.83      0.83      0.83      1409
```

Figure 53:Logistic Regression Accuracy and Classification Report

Although accuracy has traditionally been a popular evaluation metric, it only considers the proportion of correct predictions. This makes it less effective when dealing with imbalanced datasets, as it doesn't account for the model's performance on individual classes. (Kundu, 2023). Therefore, additional metrics like precision, recall, and F1-score are necessary to provide a more comprehensive evaluation, particularly in cases where the data is class-imbalanced.

Unlike accuracy, F1 score is a machine learning evaluation metric that computes the predictive skill of a model by looking at its class-wise performance. It combines precision and recall using their harmonic mean and is defined based on precision and recall scores as seen in Figure 54 and 55 (Kundu, 2023).

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Figure 54: Calculating Precision and Recall (Kundu, 2023)

$$\text{F1 Score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Figure 55: Formula of F1 score (Kundu, 2023).

The classification report as shown in Figure 53 provides deeper insight into how well the model performs by computing precision, recall and F1 score. The precision score for Class 1(churn) is 0.88, indicating that 88% of the customers predicted as churners by the model are correct. In comparison, the precision for Class 0(stay) is 0.69 showing that model is slightly less effective at predicting customers who stay.

3.5.2 Random Forest

Random Forest is a versatile and powerful supervised machine learning algorithm used for both classification and regression tasks. By combining the predictive ability of multiple tree-based models, this robust algorithm works by creating multiple decision trees on various data samples. Each tree provides a prediction, and the algorithm then selects the final prediction based on majority vote. This approach makes Random Forest robust, improving accuracy, reducing the risk of overfitting. Figure 56 illustrates a detailed representation of Random Forest Model, explaining its structure and functionality (Saini et al., 2021).

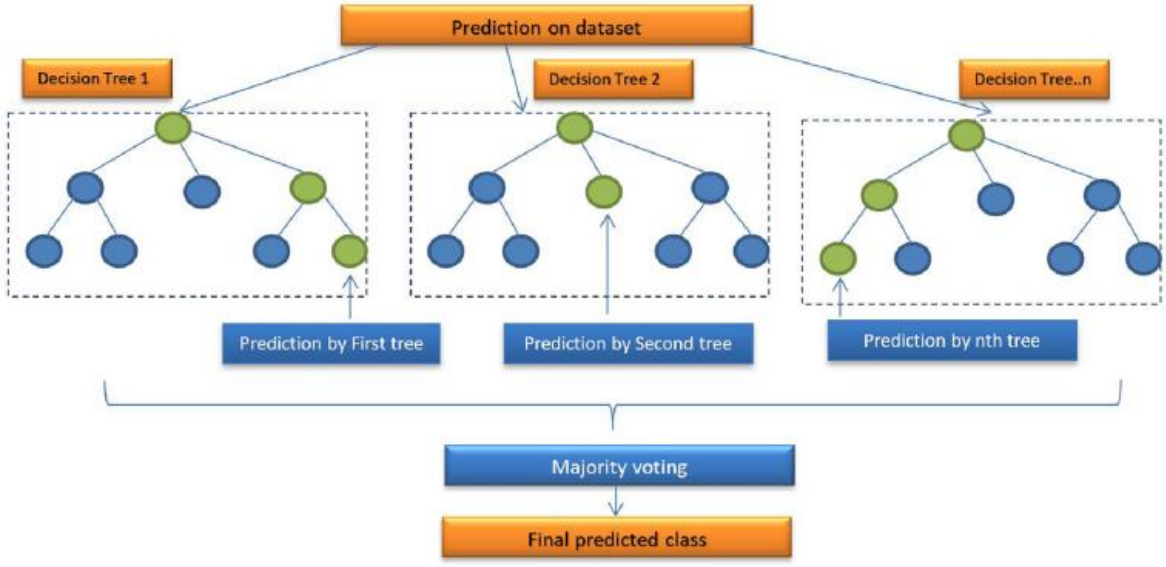


Figure 56: Cannonical Diagram of Random Forest Model (Saini et al., 2021)

3.5.2.1 Model Implementation

In this research, Random Forest was chosen for its robustness, and to avoid overfitting by averaging multiple decision trees. This was quite useful particularly with this dataset as it had both numerical and categorical variables in the dataset.

```
######################################### Random Forest Model ##############################################################
# Defining features and target variable
X = encoded_data.drop(columns=['Customer ID', 'City', 'Latitude', 'Longitude','Customer Status', 'Churn Reason','Churn Category'])
Y = encoded_data['Customer Status']

# Split the dataset into training and testing sets
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

# Standardizing the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Initializing the Random Forest model
rf = RandomForestClassifier(random_state=42)
rf.fit(X_train, Y_train)

# Make predictions on the test set
Y_pred_rf = rf.predict(X_test_scaled)
```

Figure 57: Random Forest Implementation

Figure 57 illustrates the Random Forest implementation, like Logistic Regression. The feature set ‘X’ and target variable ‘Y’ were defined in the same way as Logistic Regression. The model was built using the ‘RandomForestClassifier()’ function from scikit-learn, trained with an 80:20 train-test split, and predictions were made on the test set, with results stored in the Y_pred_rf variable.

3.5.2.2 Results and Accuracy

```
Random Forest Accuracy: 74.31%

Classification Report using Random Forest:
              precision    recall  f1-score   support

           0       0.51      0.75      0.61       373
           1       0.89      0.74      0.81      1036

    accuracy                           0.74      1409
   macro avg       0.70      0.75      0.71      1409
weighted avg       0.79      0.74      0.76      1409
```

Figure 58: Random Forest Accuracy and Classification Report

After training the Random Forest model on the pre-processed data, the performance of the model was evaluated using the testing dataset. The key metrics used to evaluate Random Forest effectiveness were the same as Logistic Regression. As shown in Figure 58, the Random Forest has achieved an overall accuracy of 74.31%. And the F1 score for Class 0(stay) observed at 0.61 is significantly lower than that for Class 1(stay) which is 0.81.

3.5.3 Comparative Analysis of Model Performances

Metric	Logistic Regression (Full Features)	Random Forest (Full Features)	Logistic Regression (Selected Features)	Random Forest (Selected Features)
Accuracy	83.32%	74.31%	82.61%	81.55%
Precision (Class 1: Churn)	0.88	0.81	0.84	0.82
Precision (Class 0: Stay)	0.69	0.61	0.59	0.68
Recall (Class 1: Churn)	0.85	0.81	0.86	0.84
Recall (Class 0: Stay)	0.73	0.63	0.55	0.66
F1-Score (Class 1: Churn)	0.86	0.81	0.85	0.83
F1-Score (Class 0: Stay)	0.71	0.71	0.57	0.67

Figure 59:Comparison of Logistic Regression and Random Forest Model Performance with Full and Selected Feature Sets (self-generated)

The table above, shown in Figure 59, compares key performance metrics—accuracy,

precision, recall, and F1-score—between the Logistic Regression and Random Forest models. It highlights the improvements in model performance following feature selection using correlation analysis and Random Forest feature importance. Both models showed enhanced performance with the selected features, though Logistic Regression consistently outperformed Random Forest, particularly in predicting customers likely to churn (Class 1).

4. Conclusion

This study sheds light on the pressing need for accurate predictions of customer churn in the telecommunications industry. By focusing on effective feature selection and rigorous model evaluation, we gained valuable insights into the factors that influence customer behaviour. Using Logistic Regression and Random Forest algorithms, we discovered key elements such as contract type, monthly charges, and tenure that play a significant role in determining whether a customer chooses to stay or leave.

Choosing the right features not only improved our model's performance but also enabled us to categorize customers into distinct groups: Prime, Mid-Tier, and High Churn Risk. This segmentation is crucial for tailoring marketing strategies and optimizing resource allocation, ultimately enhancing customer retention efforts.

Our results revealed that Logistic Regression outperformed Random Forest, achieving an impressive accuracy of 83.32% compared to 74.31% for Random Forest. This underscores the importance of selecting the right model based on the unique characteristics of the data and the specific objectives of the analysis.

In summary, the insights from this research provide a solid foundation for telecom companies looking to adopt data-driven approaches to manage customer churn. By understanding key features and distinct customer segments, businesses can develop more effective retention strategies and boost customer satisfaction. Looking ahead, exploring advanced modelling techniques and larger datasets could further refine these predictive insights, making them even more relevant in the ever-changing telecom landscape.

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